Computational Astrophysics

AA 2026-2027

Prof. Fabrizio Brighenti

1st project: hydrostatic equilibrium in a galaxy cluster/group + turbulent diffusion of metals in the hot gas

This project includes various steps and **has not a unique solution** (as most real problems in astrophysics…). **Students can and should be creative and experiment**. Talk to me for advices and guidance.

In short, we want to investigate the **structure of the gaseous halo within a galaxy cluster**. As people obsessively say: "clusters are the most massive virialized structures in the universe" and are indeed important astrophysical objects for many reasons.
Great lab to study the interaction of galaxies with the environment.

Mon. Not. R. Astron. Soc. **359,** 1041–1048 (2005) doi:10.1111/j.1365-2966.2005.08965.x

Impact of stochastic gas motions on galaxy cluster abundance profiles

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ABSTRACT

The impact of stochastic gas motions on the metal distribution in cluster cores is evaluated. Peaked abundance profiles are a characteristic feature of clusters with cool cores, and abundance peaks are probably associated with the brightest cluster galaxies (BCGs), which dwell in cluster cores. However, the width of the abundance peaks is significantly broader than the BCG light distribution, suggesting that some gas motions are transporting metals originating from within the BCG. Assuming that this process can be treated as diffusive, and using the brightest X-ray cluster A426 (Perseus) as an example, we estimate that a diffusion coefficient of the order of 2×10^{29} cm^2 s^{-1} is needed to explain the width of the observed abundance profiles. Much lower (higher) diffusion coefficients would result in too peaked (too shallow) profiles. Such diffusion could be produced by stochastic gas motions, and our analysis provides constraints on the product of their characteristic velocity and their spatial coherence scale. We speculate that the activity of the supermassive black hole of the BCG is driving the stochastic gas motions in cluster cores. When combined with the assumption that the dissipation of the same motions is a key gas heating mechanism, one can estimate both the velocity and the spatial scale of such diffusive processes.

Mon. Not. R. Astron. Soc. **372,** 1840–1850 (2006) doi:10.1111/j.1365-2966.2006.10977.x

Effect of turbulent diffusion on iron abundance profiles

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ABSTRACT

We compare the observed peaked iron abundance profiles for a small sample of groups and clusters with the predictions of a simple model involving the metal ejection from the brightest galaxy and the subsequent diffusion of metals by stochastic gas motions. Extending the analysis of Rebusco et al., we found that for five out of eight objects in the sample an effective diffusion coefficient of the order of 10^{29} cm^2 s^{-1} is needed. For AWM4, Centaurus and AWM7 the results are different suggesting substantial intermittence in the process of metal spreading across the cluster. There is no obvious dependence of the diffusion coefficient on the mass of the system.

We also estimated the characteristic velocities and the spatial scales of the gas motions needed to balance the cooling losses by the dissipation of the same gas motions. A comparison of the derived spatial scales and the sizes of observed radio bubbles inflated in the ICM by a central active galactic nucleus (AGN) suggests that the AGN/ICM interaction makes an important (if not a dominant) contribution to the gas motions in the cluster cores.

Basically, we will:

1) Find a **hydrostatic configuration** of hot gas within the potential well of a galaxy cluster
2) Study how the metals (Fe), released by the SNe in the central galaxy (BCG), diffuse in the intracluster medium (ICM) and make the gas metal-rich.
This is an important problem, with no clear solution yet.

For point 1) we need to solve the **hydrostatic equilibrium** (HE) equation:

$$\frac{dP}{dr} = -\frac{GM(<r)}{r^2}\rho$$ **which is a I order ordinary differential eq. (ODE).**

Then, we need to solve a **diffusion equation (partial differential equation, PDE),** which is a slightly more complex task.

This is just a **numerical project**, but some **astrophysical background** is needed in order to understand what we are doing (not part of the exam).

Astrophysical background

(not something I will ask during the exam)

You will learn later in several classes about "clusters", here only a schematic description will be given, the bare minimum to understand what we are doing with this project (but ask me more if you are interested).

Clusters are **systems of galaxies**, **few 1000s** for massive clusters, held together by the gravitational force mainly due to a **massive dark matter (DM) halo**. While clusters are named after the galaxies within them, **most of the baryonic matter resides in the hot gaseous halo**, invisible at optical frequencies, but shining in the **X-ray** band.

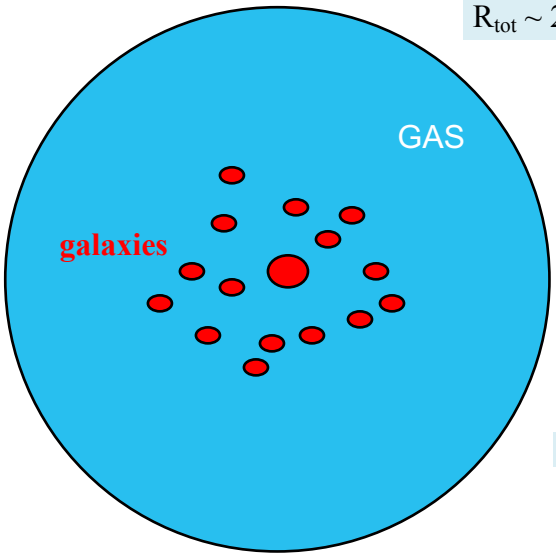
The system is **virialized**, in a dynamical equilibrium. In the same volume, of linear size **few Mpc**, galaxies, gas and DM co-exist. **DM is the most massive component (~83% of M_{tot}), followed by the gas (ICM) with ≳ 15% M_{tot}, and the galaxies (~2 % M_{tot}).**

The **interaction between galaxies, massive black holes and ICM** is a key subject in modern astrophysics, which include **many, many physical processes**. A large number of people in Bologna are involved in this research field.

What is a (massive) galaxy cluster?

$M_{tot} \sim 10^{15}\ M_\odot$

$R_{tot} \sim 2–3$ Mpc



GAS:

$T \sim 10^8$ K

$n \sim 10^{-1}$-10^{-3} cm^{-3}

$R \sim 1$ Mpc

$L_x \sim 10^{45}$ erg/s

$M_g \sim 10^{13\text{-}14}\ M_\odot$

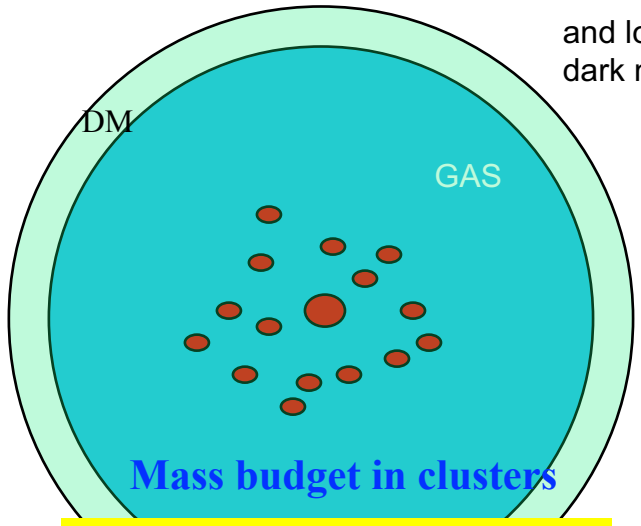
GALAXIES:

few × 1000

$M_* \sim 0.1 \times M_g$

(yes, most baryons are in gas, not stars)

Gas is **almost hydrostatic**, supported by thermal pressure. Galaxies are instead orbiting with $\sigma \sim 1000$ km/s, and the galaxy system is supported by the dynamical pressure.



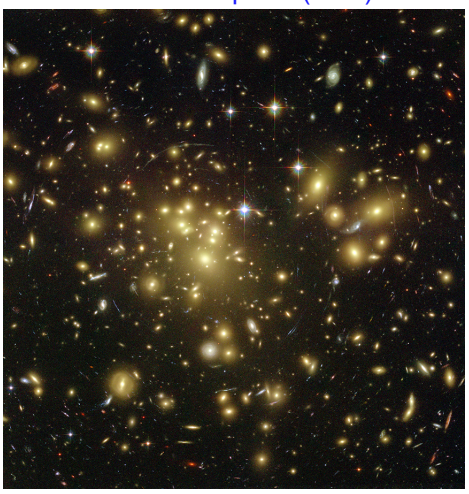
Mass budget in clusters

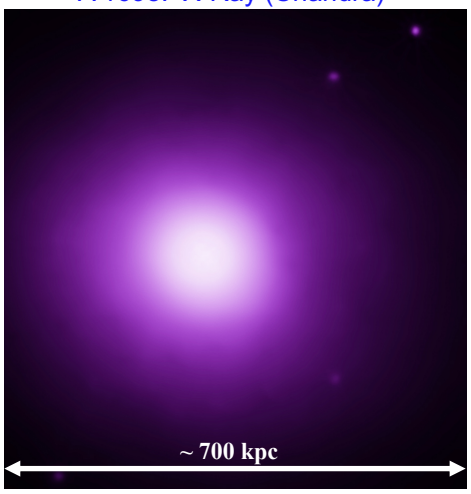
galaxies (stars!):	2%	baryons = 16 %
hot gas:	14%	
dark matter:	84%	

Galaxy Cluster at X-ray & Visual wavelengths

A 1689: Optical (HST)

A 1698: X-Ray (Chandra)





$L_x = 10^{43}$ - 10^{45} erg s^{-1} $T_{gas} = 10^7$ - 10^8 K $n_{gas} \sim 0.1 - 10^{-3}$ cm^{-3}

(bremsstrahlung, lines)

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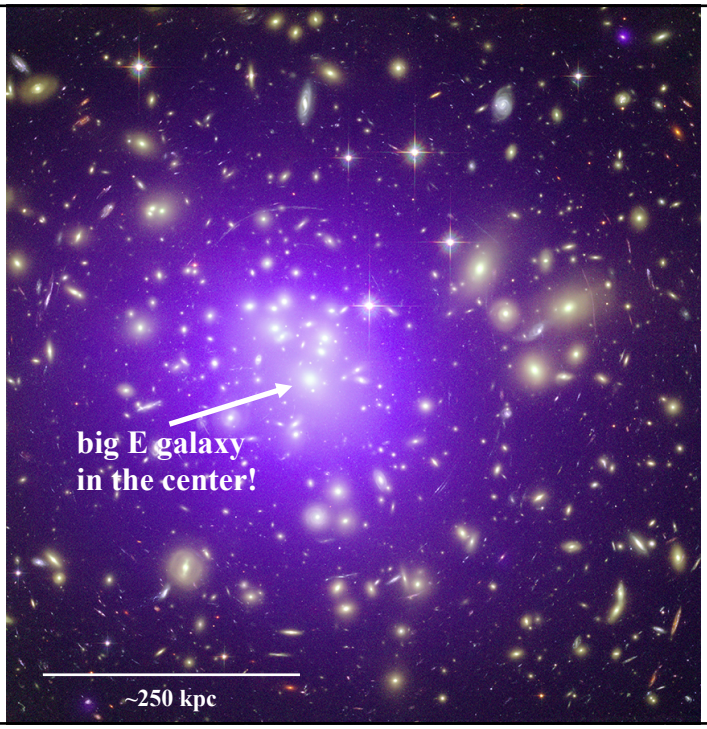
A1689

Optical
+ X-ray



Many clusters host a **large E galaxy in their center**, with $M_* \sim 5 \times 10^{11}$ - 10^{12} $M_\odot$. This galaxy is a key player for the evolution of the cluster (you'll learn more later).

For us it is important to realize that in a E galaxy explode SNIa, which inject energy (~ 10^{51} erg each) and metals, especially Fe (~ 0.7 $M_\odot$ each).

Also, the central galaxy (aka **BCG**) has a very **massive BH** at the center, which provides a **HUGE amount of energy to the ICM** (the so called **AGN feedback**) controlling the thermal evolution of the gas → **we believe that the ICM is almost in thermal equilibrium** (AGN energy injected ≈ energy radiated away) on a time average sense.

Metals in the ICM

The ICM has been **chemically enriched** by the stars (SNe, mostly) to a level comparable to the ISM in the MW (indeed, we observe mostly Fe → **iron is our metal tracer**). Usually, $Z_{Fe} \sim 0.3-0.5$ solar ($Z_{Fe,\odot} = 0.0018$). This has several crucial implications, but for the time being we will be interested only in the **origin, amount and spatial distribution of the metals.**

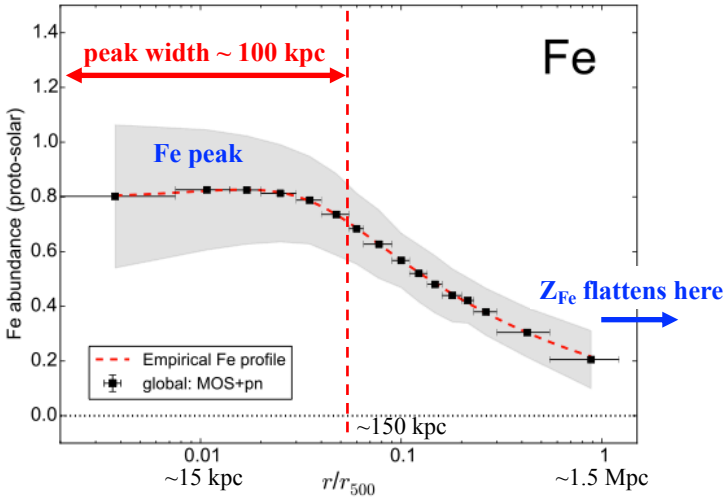
Note: $Z_{Fe,\odot} = \left(\frac{\rho_{Fe}}{\rho_H}\right)_\odot = 0.0018$ = iron solar abundance (by mass)

Observed Fe abundance in Clusters

Largest homogeneous sample: **44 CC clusters, groups and massive Es**

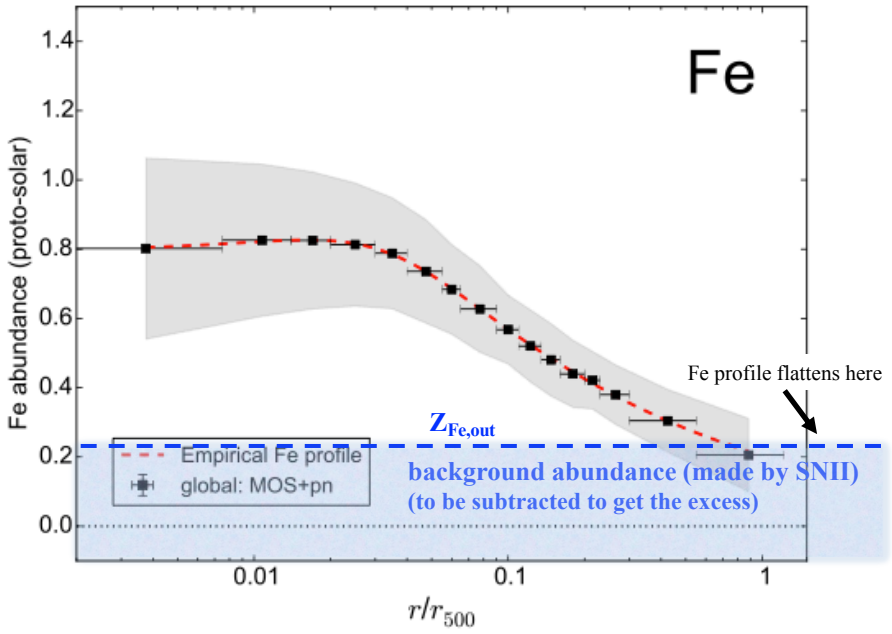
Mernier et al. 2017

average Fe abundance

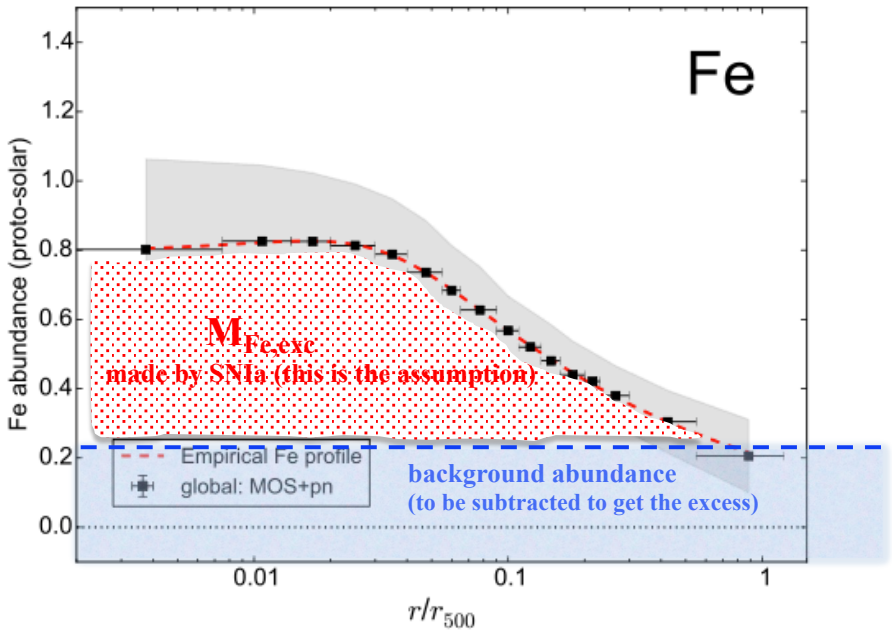


(what is r_{500}? ~ ½ R_{vir})

11



(Note: most of the cluster volume has Z_{Fe} ~ $Z_{Fe,out}$)



(Note: most of the cluster volume has $Z_{Fe} \sim Z_{Fe,out}$)

What is the problem with the Fe abundance?

$Z_{Fe,exc} = Z_{Fe} - Z_{Fe,out}$

Iron distribution much more extended than the central galaxy, supposed to have generated the Fe by SNIa. Why?

Astro questions:

1) is the amount of Fe in the peak generated by the SNIa exploding in the BCG?

2) what does spread the Fe through the ICM?

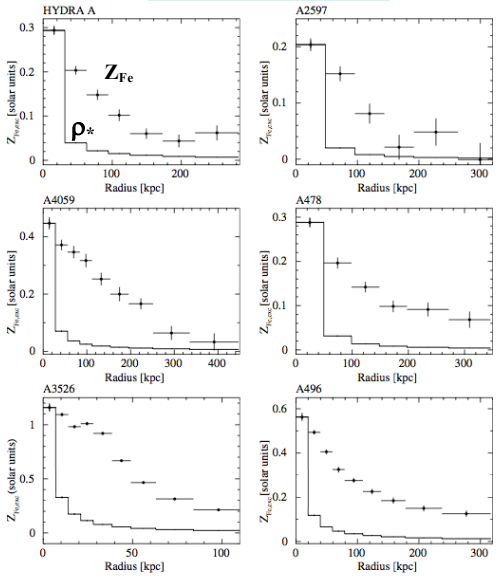


Fig. 3. Comparison between the observed (black dots) and expected (red lines) Fe abundance excess profiles for the galaxy clusters in the *XMM-Newton* local sample.

De Grandi et al, 2014)

What would we like to understand?

What is spreading the Fe outward? Or, perhaps, the Fe-peak is not created by the BCG?

Several researchers think that **turbulence is mixing the gas**, so diffusing the metals outward. This seems reasonable, turbulence must be present at some level.
This can be studied by solving a simple PDE (as a first approximation).

But **first** we need to understand how the gas is distributed in the cluster, and this can be calculated **by assuming hydrostatic equilibrium** and a model for the underlying potential well.

$$\frac{dP}{dr} = -\frac{GM(<r)}{r^2}\rho$$

So, let's start to set the problem.
What is the gravitational model appropriated for a group/cluster?

Hot gas is in equilibrium in the gravitational potential of the cluster, which is dominated by the dark matter. Thus, the first step is to set the DM distribution and gravity.

DM density profile: theory and observations suggest that DM halos have a density profile of the form:

$$\rho_{dm}(r) = \frac{\rho_{dm,0}}{\left(\frac{r}{r_s}\right)\left(1+\frac{r}{r_s}\right)^2}$$

(NFW profile, Navarro, Frenk & White, 1996)

The parameters $\rho_{dm,0}$ and r_s depend on the **cluster mass**, and in our case with $M_{virial} \sim M_{DM} \sim 1.2 \times 10^{15}\ M_\odot$, are:

$$\rho_{dm,0} = 7.35 \times 10^{-26}\ \mathrm{g/cm^3} \qquad r_s = 435.7\ \mathrm{kpc}$$

The **virial radius** of this cluster is $r_{vir} \sim 2.8$ Mpc.

So, we know the DM density profile → we can calculate the **DM mass profile**, $M_{DM}(r)$ by integration.

Analitically: $M_{DM}(r) = \int_0^r \rho_{DM}(r) dV = \int_0^r 4\pi r^2 \rho_{DM}(r) dr$

Indeed the integration can be carried out analitically (see below), but since we are here to learn how to use numerical methods, let's the computer do it.

The basic idea is to represent all the quantities (mass, density, etc.) on a **discrete grid of points**, rather than on the continuum space

→ **this transforms the derivatives in difference ratios**

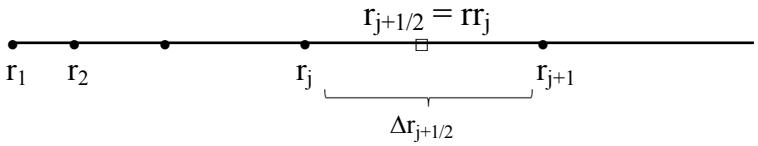
→ **the differential equation becomes an algebraic equations.**

Because we are assuming spherical symmetry, **these points represents the radial coordinates of spherical shells.**

I step: build the **numerical grid(s)** (Why **two staggered grids**?)

- uniform or not? (maybe more refined at the center?)

  for instance: $\Delta r_{j+1/2} = f \cdot \Delta r_{j-1/2}$ con $f > 1$)
- $r_{min} = 0.1$ kpc; $r_{max} = 3$ Mpc
- check the results (always)!



$$r_{j+1} = r_j + f \cdot \Delta r_{j-1/2} = r_j + f \cdot (r_j - r_{j-1})$$

$$r_{j+1/2} = rr_j = \frac{1}{2}(r_j + r_{j+1})$$

In spherical coordinates, the **volume between two shells**, of radii r_j and r_{j+1} is

$$\Delta V_{j+1/2} = \frac{4}{3}\pi(r_{j+1}^3 - r_j^3)$$

Or just use a **uniform** grid (preferred option for now):

```
!    set the grid

rmin = 0.*cmkpc
rmax = 3000.*cmkpc
do j=1,jmax
   r(j)=rmin+(j-1)*rmax/(jmax-1)
enddo
do j=1,jmax-1
   rr(j)=r(j)+0.5*(r(j+1)-r(j))
enddo
rr(jmax)=rr(jmax-1)+(rr(jmax-1)-rr(jmax-2))
open(10,file='grid.dat',status='unknown')
do j=1,jmax
   write(10,*)real(r(j)/cmkpc),real(rr(j)/cmkpc)
enddo
close(10)
```

program heq.f90 on Virtuale (not anymore…)

II step: integrate ρ_{DM} to get $M_{DM}(<r)$ (needed for the gravity)

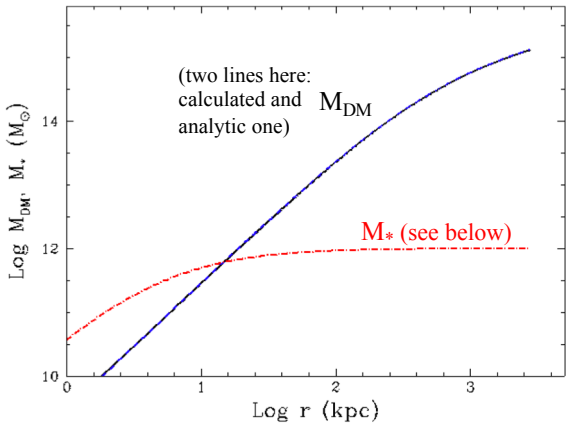
That is, you need to **sum the DM mass over every shell**, the DM mass in the shell at "j+1/2" being

$$\Delta M_{DM,j+1/2} = \rho_{DM,j+1/2} \Delta V_{j+1/2} = \rho_{DM,j+1/2} \frac{4}{3}\pi(r_{j+1}^3 - r_j^3)$$

You should get this:

I plot you have to include in the report!

Note the axes units, etc.



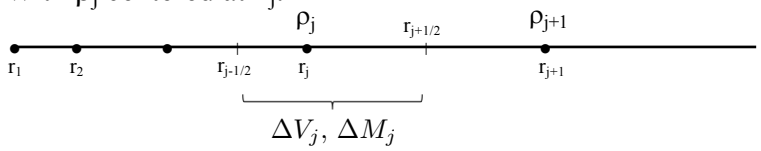
Check the numerical mass profile $M_{dm}(r)$ against the **analytical solution**:

$$M_{dm}(r) = 4\pi\rho_{dm,0}r_s^3\left[ln\left(1+\frac{r}{r_s}\right)-\frac{r}{r+r_s}\right]$$

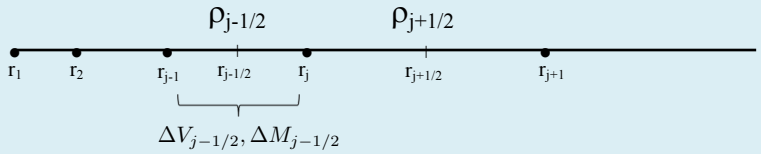
Overplot this function to the one previously calculated: they should be almost identical.

N.B. it is **important where we center the variables**. For instance, let's center the density at $r_{j+1/2}$ (that is, ρ_{DM} and ρ_j are calculated at rr_j). **Then ΔV and ΔM must be calculated consistently.**

1) With ρ_j centered at x_j:



2) Or, with $\rho_{j+1/2}$ centered at $x_{j+1/2}$: ← **the code uses this**



```
vol(1)=4.1888*r(1)**3
do j=2,jmax   ←
   vol(j)=4.1888*(r(j)**3-r(j-1)**3)
enddo
```

!! centered at $r_{j\text{-}1/2} = rr(j\text{-}1)$

```
open(20,file='masse.dat')
mnfw(1)=0.
do j=2,jmax
   x=r(j)/rs
   mnfw(j)=mnfw(j-1)+rhonfw(j-1)*vol(j)   ← calculated DM mass
   mdark(j)=mvir*(log(1.+x)-x/(1.+x))/fc  ← analytic DM mass
   mhern(j)=mbcg*r(j)**2/(r(j)+ahern)**2  ← analytic stellar mass
   write(20,1001)r(j)/cmkpc,mnfw(j)/msol,mdark(j)/msol,mhern(j)/msol
enddo
1001 format(4(1pe12.4))
close(20)
```

Now that you have $M_{DM}(r)$ we can proceed to the **next step:** integrate the **hydrostatic equilibrium equation**.

Consider **first an isothermal gas**: T= 7.67 keV = 8.9 × 10^7 K (1 keV = 1.16×10^7 K)

The equation to solve is:

$$\frac{dP}{dr} = -\frac{GM(<r)}{r^2}\rho_g \qquad \text{where} \quad P = \frac{K\rho_g T}{\mu m_p} \qquad \mu = 0.61$$

($K = 1.38 \times 10^{-16}$ (cgs))

If the gas is **isothermal** the equation for the **gas density** is:

$$\frac{d\ln\rho_g}{dr} = -\frac{GM(<r)}{r^2}\frac{\mu m_p}{KT}$$

why ln ρ?

This equation requires the **initial condition $\rho_g(0)$**. We choose it by **requiring that the baryon fraction** of the cluster, $f_b = M_{bar}/M_{tot} \sim 0.16$, which is the cosmological value.
→ **the I.C. depends on the numerical grid you use, so you have to experiment.**

With my grid: $\rho_g(0) = \rho_0 = 4 \times 10^{-26}$ g/cm^3 (this is the central **gas density**!)

The above ODE transforms in a FDE (let $\rho_g = \rho$):

$$\frac{d\ln\rho}{dr} \rightarrow \frac{\ln\rho_{j+1/2} - \ln\rho_{j-1/2}}{r_{j+1/2} - r_{j-1/2}}$$

derivative is centered in "j"

this difference is centered in r_j → the mass on the r.h.s. must be M_j,

→ the recursive equation to solve for ρ is:

$$\ln\rho_{j+1/2} = \ln\rho_{j-1/2} - \Delta r \frac{GM_j}{r_j^2}\frac{\mu m_p}{kT}$$

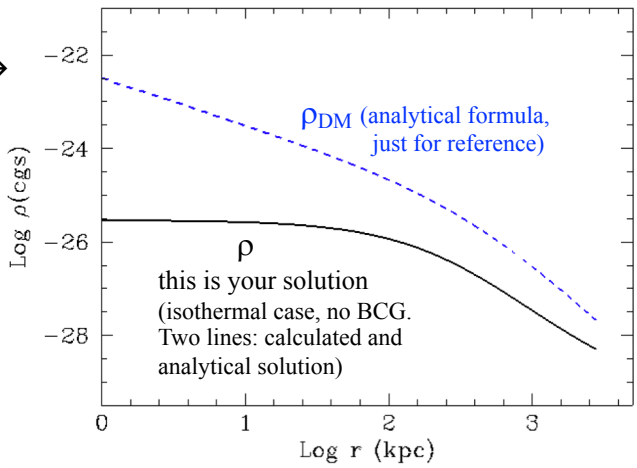
Check the centering of the variables.
It is important to make everything consistent to get an accurate solution.

III step: integrate the hydrostatic equilibrium equation

$$\ln\rho_{j+1/2} = \ln\rho_{j-1/2} - \Delta r \frac{GM_j}{r_j^2}\frac{\mu m_p}{kT}$$

Start at j = 1, where $\rho_1 = \rho_g(0)$, and proceed to j = jmax.

You'll get this →



II plot you have to include in the report!

Check the numerical solution for ρ(r) with the **analytic** one:

$$\rho(r) = \rho_0 \exp\left\{-\frac{27}{2}\, b \left[1 - \frac{\ln(1 + r/r_s)}{r/r_s}\right]\right\}$$
$$= \rho_0 \mathrm{e}^{-27b/2} \left(1 + \frac{r}{r_s}\right)^{27b/(2r/r_s)}$$

Notice: ρ is indeed defined on $r_{j+1/2}$!

with: $$b = \frac{8\pi G \mu m_p \rho_{dm,0} r_s^2}{27kT}$$

Last check: calculate the "**baryon fraction**" f_b(r) at R_{vir}:

$$f_b = \frac{M_b}{M_{tot}} = \frac{M_*(r) + M_g(r)}{M_{DM}(r) + M_*(r) + M_g(r)}$$

and verify that it approaches the **cosmic value ~ 0.16** at the virial radius of the cluster (~2.8 Mpc). Otherwise, change the initial condition for ρ appropriately.

Now make it more realistic: add the **central elliptical galaxy (BCG)**

$$M_*(r) = M_{BCG} \frac{r^2}{(r+a)^2}$$

Hernquist profile (1990) for the BCG

$$r_{1/2} = (1 + \sqrt{2})a = 12 \ \mathrm{kpc}$$

(half mass radius)

$$M_{BGC} = 10^{12} \ \mathrm{M}_\odot$$

N.B. this is the **total stellar mass** of the central galaxy

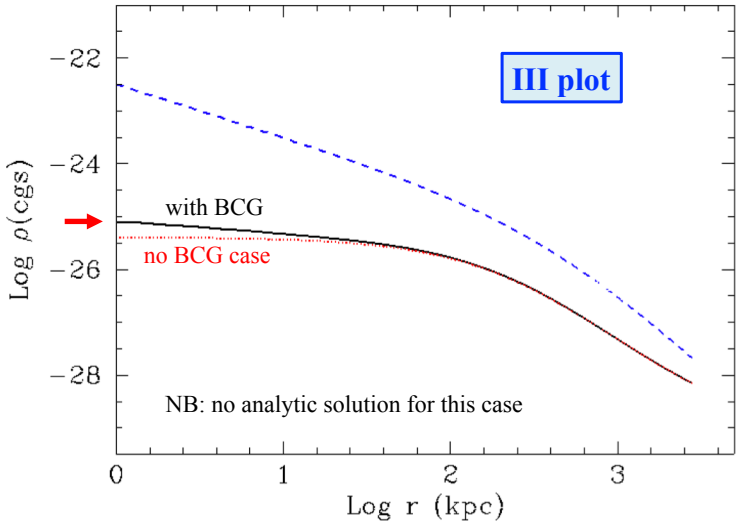
$$\phi_{BCG}(r) = -\frac{GM_{BCG}}{r+a}$$

Thus, the new total mass is $M_{DM}(r) + M_*(r)$ → apply the same procedure as before

Last step: change ρ_0 (central gas density) to get: $(M_{gas} + M_*)/M_{total} \sim 0.16$

→ The extra gravity provided by the BCG steepens somewhat the density profile (why?) where the BCG potential is significant, that is in the central region.

→ This is why **you need to increase ρ_0 to have a baryon fraction ~ 0.16**.



Not finished yet! Consider now a more realistic T profile

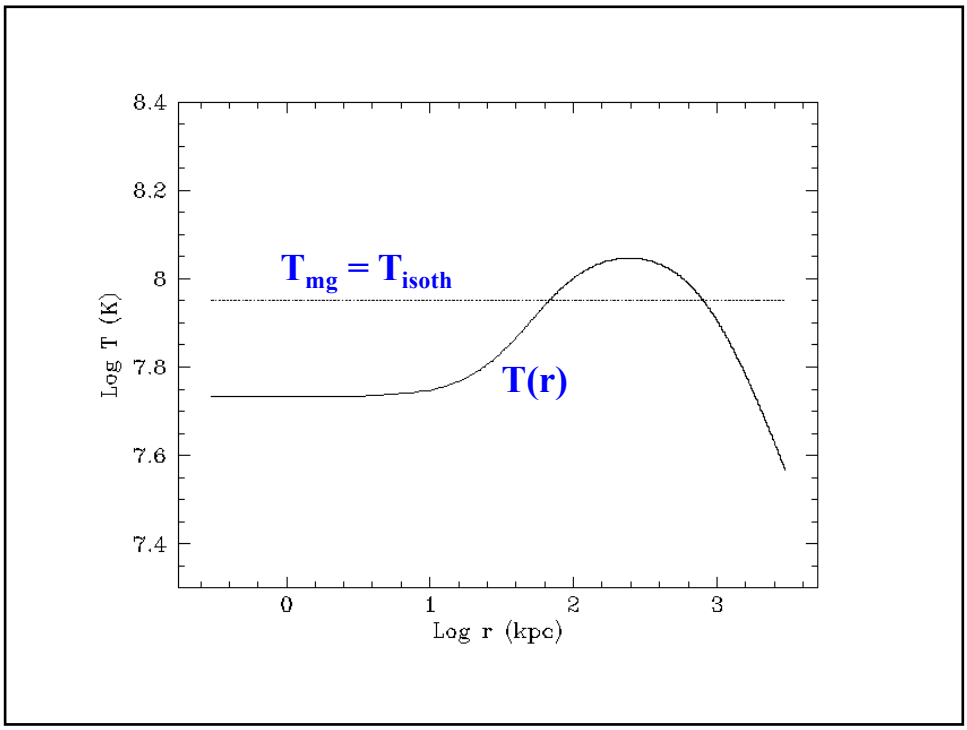
$$\frac{T(r)}{T_{mg}} = 1.35\,\frac{(x/0.045)^{1.9}+0.45}{(x/0.045)^{1.9}+1}\,\frac{1}{(1+(x/0.6)^2)^{0.45}}$$

where $x = r/r_{500}$ $\quad r_{500} \sim r_{vir}/2 = 1.4$ Mpc

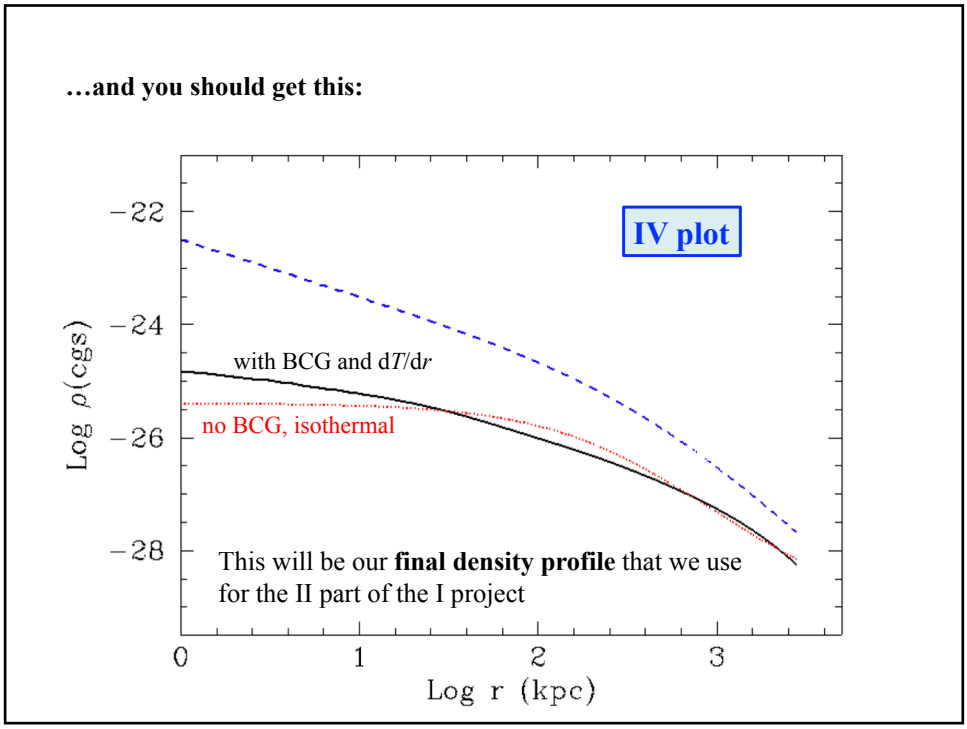
$T_{mg} = 8.9 \times 10^7$ K

Use the correct hydrostatic equilibrium equation when T is not constant! T is centered like ρ at $r_{j+1/2}$

Again, you need to find ρ_0 in order to have $f_b \sim 0.16$



…and you should get this:



```
!*********************************************************************
!  this program solves the hydrostatic equilibrium equation
!  for an isothermal gas in a NFW halo
!*********************************************************************

parameter(jmax=5000)
implicit real*8 (a-h,o-z)
real*8, dimension(jmax) :: r(jmax),rr(jmax),vol(jmax),mnfw(jmax),&
        rhost(jmax),rho(jmax),mhern(jmax),rhonfw(jmax),mdark(jmax),&
        grvnfw(jmax),lnd(jmax)
real*8 :: msol,mu,mp,rmin,rmax,mvir,rvir,mbgc,ahern

!  constants

msol = 1.989d33
cmkpc = 3.084e21
mu=0.62
boltz=1.38066e-16
guniv=6.6720e-8
mp=1.67265e-24
```

See program `heq.f90` on the Virtuale website
(I apologize for some comment in Italian…)
Oops! Not anymore…

```
!    set the grid

rmin = 0.*cmkpc
rmax = 3000.*cmkpc
do j=1,jmax
   r(j)=rmin+(j-1)*rmax/(jmax-1)
enddo
do j=1,jmax-1
   rr(j)=r(j)+0.5*(r(j+1)-r(j))
enddo
rr(jmax)=rr(jmax-1)+(rr(jmax-1)-rr(jmax-2))
open(10,file='grid.dat',status='unknown')
do j=1,jmax
   write(10,*)real(r(j)/cmkpc),real(rr(j)/cmkpc)
enddo
close(10)

vol(1)=4.1888*r(1)**3
do j=2,jmax
   vol(j)=4.1888*(r(j)**3-r(j-1)**3)
enddo
```

This is a uniform grid!

!! the "rr" grid is shifted by half Δr: $r_{j+1/2}$

!! centrato a rr(j-1) !!

```
!  parametri del problema

rho0nfw=7.35d-26
rs=435.7*cmkpc
rho0=2.882d-26
ticm=8.9e7

rvir=2797.*cmkpc
fc=1.138799
mvir=1.3e15*msol
mbgc=1.d12*msol
ahern=10.*cmkpc/(1.+2.**(0.5))

do j=1,jmax
   x=rr(j)/rs
   rhonfw(j)=rho0nfw/(x*(1.+x)**2)
enddo
```

← This is ρ_0 and must be changed

← ρ_{NFW} is centered ar $r_{j+1/2}$

```
open(20,file='masse.dat')
mnfw(1)=0.
do j=2,jmax
   x=r(j)/rs
   mnfw(j)=mnfw(j-1)+rhonfw(j-1)*vol(j)
   mdark(j)=mvir*(log(1.+x)-x/(1.+x))/fc     !! analytic formula, just for checking
   mhern(j)=mbgc*r(j)**2/(r(j)+ahern)**2
   write(20,1001)r(j)/cmkpc,mnfw(j)/msol,mdark(j)/msol,mhern(j)/msol
enddo
1001 format(4(1pe12.4))
close(20)

open(20,file='grv.dat')
grvnfw(1)=0.
do j=2,jmax
   grvnfw(j)=guniv*(mnfw(j)+mhern(j))/r(j)**2   !! this is the grav acceleration, in this case
   write(20,1002)r(j)/cmkpc,grvnfw(j)/msol      !! includes both DM and stars
enddo
1002 format(2(1pe12.4))
close(20)
```

```
!     calculate the gas density, assuming ticm

lnd(1)=log(rho0)
do j=2,jmax
   gg=grvnfw(j)
   lnd(j)=lnd(j-1)-gg*(mu*mp)*(rr(j)-rr(j-1))/(boltz*ticm)
enddo

do j=1,jmax
   rho(j)=exp(lnd(j))
enddo

open(20,file='density.dat',status='unknown')
do j=1,jmax
   write(20,1000)rr(j)/cmkpc,rho(j),rhonfw(j)
enddo
close(20)
1000 format(3(1pe12.4))

stop
end
```

Complication: turbulence pressure

$$\cancel{\frac{\partial(\rho v)}{\partial t}} + \frac{\partial(\rho v^2)}{\partial x} + \underbrace{\frac{\partial p}{\partial x}}_{\text{HSE}} = -g\rho$$

can be $\neq 0$ if turbulence and/or
some bulk motion are present
→ turbulent pressure (gradient of), etc.

$$p_{\rm tot} = p_{\rm th} + p_{\rm nt}$$
$$= p_{\rm th} + \rho\langle v^2 \rangle$$

(compare with thermal pressure: $p_{\rm th} = \frac{1}{3} n m v_{\rm th}^2 = \frac{1}{3}\rho v_{\rm th}^2$)

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Addendum (not mandatory, just an extra)

We are assuming that there is **turbulence**, which means random motion (stirring) at any scale. The ICM is in time averaged equilibrium, though, with no coherent (bulk) motion present (no global inflow, outflow…).
Turbulence dynamically behaves as a **non-thermal pressure**! Need to include it in the hydrostatic equilibrium equation.

$$\frac{dP_{\rm tot}}{dr} = \frac{d(P + P_{\rm nt})}{dr} = -\frac{GM(<r)}{r^2}\rho$$

with $$P_{\rm nt} \equiv P_{\rm turb} \approx \frac{1}{3}\rho v_t^2$$

where v_t is the ‘‘characteristic’’ turbulent velocity, in the range [150 – 400] km/s.

The point is that the **turbulent pressure provides an extra-support** against the gravity → it lowers the density gradient needed to keep the gas in equilibrium.

How to include the turbulent pressure in the HSE?

$$\frac{dP_{\rm tot}}{dr} = \frac{d(P + P_{\rm nt})}{dr} = -\frac{GM(<r)}{r^2}\rho$$

$$\downarrow$$

$$\frac{d\ln\rho}{dr} = -\frac{GM(<r)}{r^2}\frac{\mu m_p}{kT} - \frac{d\ln T}{dr} - \frac{P_{\rm nt}(r)}{P(r)}\frac{d\ln P_{\rm nt}}{dr}$$

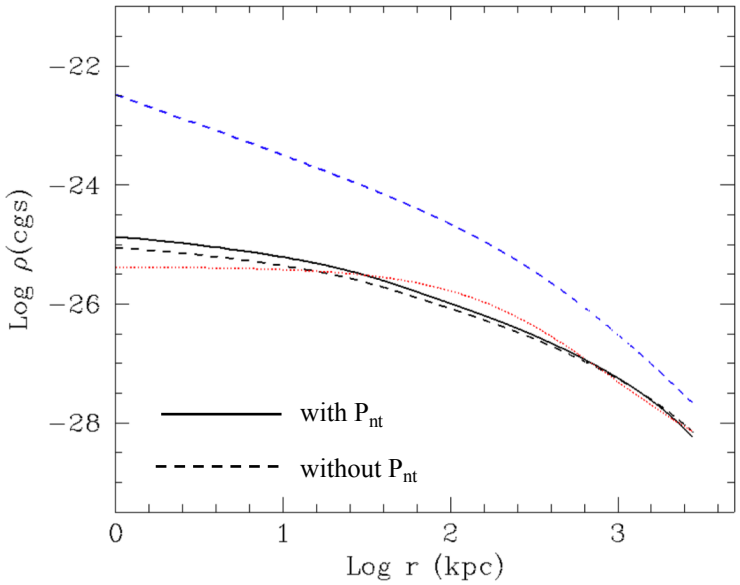
Because P_{nt} **depends on the density**: $P_{\rm nt} \equiv P_{\rm turb} \approx \frac{1}{3}\rho v_t^2$, the numerical solution is more complicated. What to do?

Ex: first calculate $\rho_0(r)$ with P only, then recalculate the “real” density $\rho(r)$ with the turbulent term $1/3\rho_0 v_t^2$.

Do you expect a large variation with respect to the no-turbulence case?

Example: model with BCG and dT/dr, v_{turb} = 500 km/s (constant)
(3 iterations)

Large v_t!!



Qualitative effects of turbulent pressure on the density profile.

The equation can also be written as:

$$\frac{d\ln\rho}{d\ln r} = -\frac{GM(<r)}{r}\frac{\mu m_p}{kT} - \frac{d\ln T}{d\ln r} - \frac{P_{\rm nt}(r)}{P(r)}\frac{d\ln P_{\rm nt}}{d\ln r}$$

which relates the slope of the density profile with the slopes of the other quantities of the problem.

It is particularly instructive the simple **isothermal case** and with a **spatially constant P_{nt}/P ratio**. The equation becomes:

$$\frac{d\ln\rho}{d\ln r} = \underbrace{-\frac{GM(<r)}{r}\frac{\mu m_p}{kT}}_{\text{value depends on the M/r profile}} \underbrace{-\frac{P_{\rm nt}}{P}\frac{d\ln P_{\rm nt}}{d\ln r}}_{>0}$$

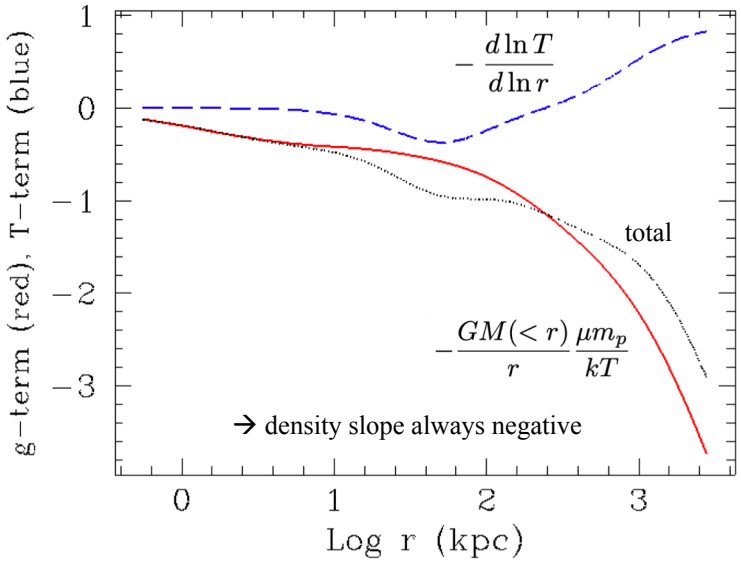
slope negative, varies with r, P_{nt} makes density flatter.

$$M(r) \propto \int r^2\rho(r)dr \propto r^2 - r - \ln r \qquad \frac{M(r)}{r} \propto r - \text{const} - \frac{\ln r}{r}$$

You can investigate this case by checking the radial profile of all the terms in the equation. You can esperiment by varying the turbulence pressure (also by making it not costant), etc.

Can you make the density profile not monotonic? What would happen In this case?

Ex: simple case (no P_{turb}): $\frac{d\ln\rho}{d\ln r} = -\frac{GM(<r)}{r}\frac{\mu m_p}{kT} - \frac{d\ln T}{d\ln r}$



Part II: turbulent diffusion of Fe in the ICM

Ref. Rebusco et al, 2005, 2006 ← **read one of these!**

Let's choose **Perseus as a model for a typical massive cluster** (that is, the cluster of the first part of the project → same density profile from hydrostatic equilibrium)

The core of the problem is to solve a **diffusion equation** for the iron density ρ_{Fe}

$$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial (r^2 D \rho \frac{\partial Z_{Fe}}{\partial r})}{\partial r} + S_{Fe}(r,t) \qquad Z_{\rm Fe} = \frac{1}{1.4}\frac{\rho_{\rm Fe}}{\rho}$$

Compare with Rebusco et al. 2005: $\frac{\partial na}{\partial t} = \nabla \cdot (Dn\nabla a) + S$

(→ same equation as above)

This equation needs:

1) the total gas density ρ(r) (known from the hydrost. equilibrium)
2) the diffusion coefficient D [possibly D(r)];
3) the initial abundance Z_{Fe}(r) (or ρ_{Fe}(r), which is the same)
4) the Fe source term $S_{Fe}(r,t)$ [if any]

The equation which describes the Fe diffusion is:

$$\frac{\partial na}{\partial t} = \nabla \cdot (Dn\nabla a) + S, \qquad \text{(7)}$$

I use the notation *a(r) = Z_{Fe}(r)*

where $n = n(r)$ is the gas density, $a = a(r, t)$ is the iron abundance $S = S(r, t)$ is the source term due to the iron injection from the central galaxy, and D is the diffusion coefficient.

Note that in the idealized case with D = ***constant*** and n = ***constant***, this becomes the **standard diffusion equation** of the form:

$$\frac{\partial f}{\partial t} = D\frac{\partial^2 f}{\partial x^2} \qquad \text{with} \qquad f = na = \rho_{Fe}$$

The **turbulent diffusion coefficient** is of the order of $D \approx v_t l_t$

with v_t ~ typical **turbulent velocity** and l_t is the typical **turbulence length-scale** (poorly defined)

Remember: parabolic PDE: diffusion

$$\frac{\partial f}{\partial t} - D\frac{\partial^2 f}{\partial x^2} = 0$$

- D = diffusion coefficient
- **represents diffusion**, mixing, thermal conduction…
- needs initial conditions f(x, t=0)

→ **numerical problems: stability, efficiency**

A preview (for our Fe-project): linear diffusion equation

The eq. to be solved is equivalent to this: $\frac{\partial f}{\partial t} = D\frac{\partial^2 f}{\partial x^2}$ if D = constant

What method should we use?
We could be tempted to reject the FTCS method for stability reasons.
That would be wrong! **While FTCS is unstable for transport problems, it is perfectly ok for diffusion equations**. Why is that? Again, think to the physics of the process.
Diffusion is isotropic, so it makes sense to use a centered spatial derivative!

Instead, FTCS applied to the advection equation does not capture the directional property of the solution. Thus:

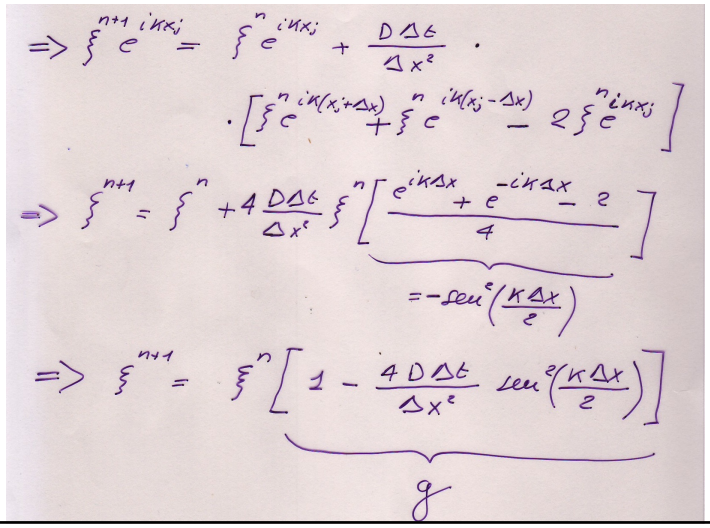
$$f_j^{n+1} = f_j^n + \frac{D\Delta t}{\Delta x^2}(f_{j+1}^n - 2f_j^n + f_{j-1}^n)$$ explicit FTCS

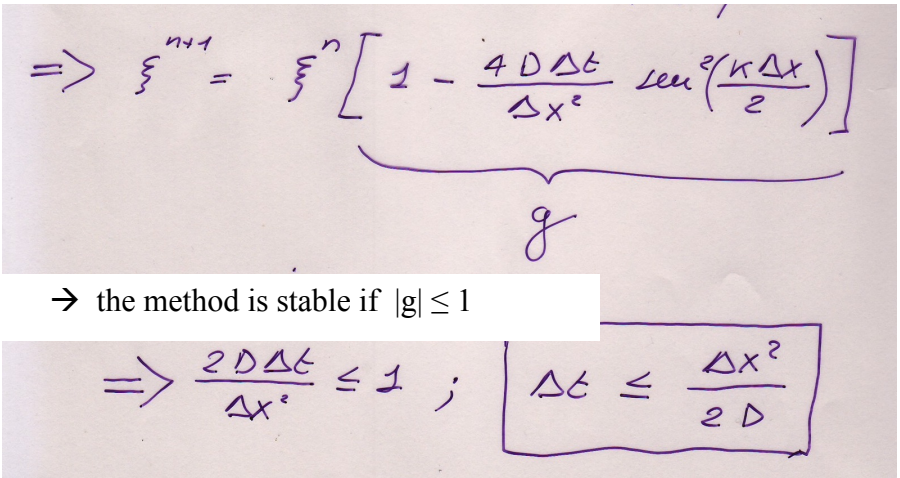
$$f_j^{n+1} = f_j^n + \frac{D\Delta t}{\Delta x^2}(f_{j+1}^{n+1} - 2f_j^{n+1} + f_{j-1}^{n+1})$$ implicit BTCS

We find that for the explicit FTCS the stability condition is: $\Delta t \leq \frac{\Delta x^2}{2D}$

Von Neumann stability analysis

As before, insert $\xi^n e^{ikx_j}$ in the FDE:





→ the method is stable if $|g| \leq 1$

$$\Rightarrow \frac{2D\Delta t}{\Delta x^2} \leq 1 \; ; \quad \boxed{\Delta t \leq \frac{\Delta x^2}{2D}}$$

Note: to investigate a problem with characteristic scalelength L, we need to calculate at least a number of timesteps (cycles):

$$N \sim \frac{L^2}{D\,\Delta t} \sim \frac{L^2}{\Delta x^2}$$

→ possible efficiency problem

Mon. Not. R. Astron. Soc. **372,** 1840–1850 (2006) doi:10.1111/j.1365-2966.2006.10977.x

Effect of turbulent diffusion on iron abundance profiles

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$$D \sim \frac{v_t l_t}{3}$$

ABSTRACT

We compare the observed peaked iron abundance profiles for a small sample of groups and clusters with the predictions of a simple model involving the metal ejection from the brightest galaxy and the subsequent diffusion of metals by stochastic gas motions. Extending the analysis of Rebusco et al., we found that for five out of eight objects in the sample an effective diffusion coefficient of the order of 10^{29} cm^2 s^{-1} is needed. For AWM4, Centaurus and AWM7 the results are different suggesting substantial intermittence in the process of metal spreading across the cluster. There is no obvious dependence of the diffusion coefficient on the mass of the system.

We also estimated the characteristic velocities and the spatial scales of the gas motions needed to balance the cooling losses by the dissipation of the same gas motions. A comparison of the derived spatial scales and the sizes of observed radio bubbles inflated in the ICM by a central active galactic nucleus (AGN) suggests that the AGN/ICM interaction makes an important (if not a dominant) contribution to the gas motions in the cluster cores.

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Impact of stochastic gas motions on galaxy cluster abundance profiles

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analysis for Perseus only

ABSTRACT

The impact of stochastic gas motions on the metal distribution in cluster cores is evaluated. Peaked abundance profiles are a characteristic feature of clusters with cool cores, and abundance peaks are probably associated with the brightest cluster galaxies (BCGs), which dwell in cluster cores. However, the width of the abundance peaks is significantly broader than the BCG light distribution, suggesting that some gas motions are transporting metals originating from within the BCG. Assuming that this process can be treated as diffusive, and using the brightest X-ray cluster A426 (Perseus) as an example, we estimate that a diffusion coefficient of the order of 2 $\times$ 10^{29} cm^2 s^{-1} is needed to explain the width of the observed abundance profiles. Much lower (higher) diffusion coefficients would result in too peaked (too shallow) profiles. Such diffusion could be produced by stochastic gas motions, and our analysis provides constraints on the product of their characteristic velocity and their spatial coherence scale. We speculate that the activity of the supermassive black hole of the BCG is driving the stochastic gas motions in cluster cores. When combined with the assumption that the dissipation of the same motions is a key gas heating mechanism, one can estimate both the velocity and the spatial scale of such diffusive processes.

Turbulent diffusion of Fe in the ICM

Ref. Rebusco et al, 2005, 2006 ← **read one of these!**

Let's choose **Perseus ("our cluster") as a model for a typical massive cluster**.
Let's also assume the **observed abundance profile** quoted in Rebusco et al. 2005: (? Talk!)

Observations

For the abundance profile $a(r)$ (Fig. 1), we use the deprojected profile obtained from *XMM–Newton* data (Churazov et al. 2003). The normalization was reduced by ~15 per cent to match better the most recent deep *Chandra* observations of the Perseus core (Schmidt et al. 2002; Sanders et al. 2004); namely,

$$a(r) = 0.3 \times \frac{2.2 + (r/80)^3}{1 + (r/80)^3} \; a_\odot, \qquad (3)$$

here $a(r) = Z_{Fe}(r)$
(the 1.4 factor is missing here!)

where r is in kiloparsecs and $a_\odot$ is the solar abundance (Anders & Grevesse 1989). The functional form used completely

!! →

$$a(r) = Z_{\rm Fe} = 1.4\frac{\rho_{\rm Fe}}{\rho} \qquad a_\odot = Z_{\rm Fe,\odot} = 1.8 \times 10^{-3}$$

Let us assume the previously calculated gas density profile in hydrostatic equilibrium (model with the BCG and $dT/dr \neq 0$).

Note that Rebusco et al. 2005 **assume as density and T profiles** the following:

The electron density n_e and temperature T_e profiles used here are based on the deprojected *XMM–Newton* data (Churazov et al. 2003, 2004), which are also in broad agreement with the *ASCA* (Allen & Fabian 1998), *BeppoSAX* (De Grandi & Molendi 2001, 2002) and *Chandra* (Schmidt et al. 2002; Sanders et al. 2004) data; namely,

$$n_e = \frac{4.6 \times 10^{-2}}{[1 + (r/57)^2]^{1.8}} + \frac{4.8 \times 10^{-3}}{[1 + (r/200)^2]^{0.87}} \; {\rm cm}^{-3} \qquad (1)$$

and

$$T_e = 7 \times \frac{1 + (r/71)^3}{2.3 + (r/71)^3} \; {\rm keV}, \qquad (2)$$

where r is measured in kiloparsecs. The hydrogen number density is assumed to be related to the electron number density as $n_{\rm H} = n_e/1.2$.

n_e electron number density

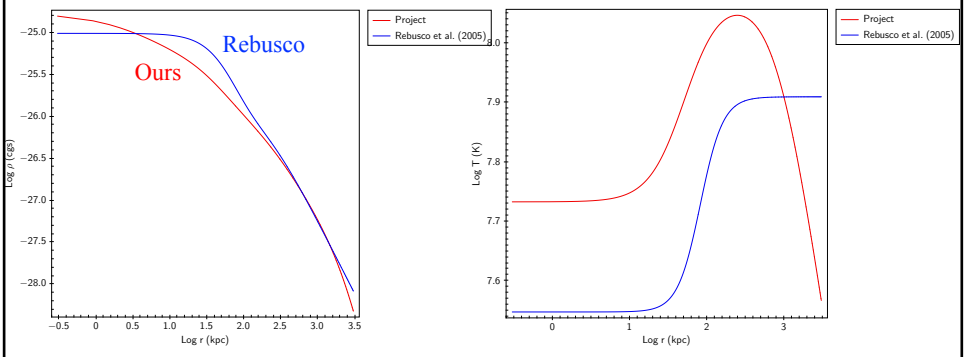
$n_{tot} \sim 1.83 n_e$

$$\rho = \mu_e m_p n_e = 1.937 \times 10^{-24} n_e \;\; {\rm g/cm^3}$$

$$1 \;\; {\rm keV} = 1.16 \times 10^7 \;\; {\rm K}$$

You can (should!) overplot them to our profiles and check the difference.

Comparison between "observed" and calculated profiles



Differences are tolerable…

Back to the equation to solve: $\frac{\partial na}{\partial t} = \nabla \cdot (Dn\nabla a) + S$

The turbulent diffusion equation can be written as an **equation for ρ_{Fe}**:

$$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial (r^2 D\rho \frac{\partial Z_{Fe}}{\partial r})}{\partial r} + S_{Fe}(r,t)$$

$$Z_{Fe} = 1.4\frac{\rho_{Fe}}{\rho}$$

This is a PDE for ρ_{Fe}(r,t), which requires I.C., that is ρ_{Fe}(r, t=0)

We need the **initial conditions** (at $t = t_{initial}$) for ρ_{Fe}(r) or Z_{Fe}(r), and
1) we can use the fit of the **observations** quoted in Rebusco et al. 2005.
2) or we can **start with $Z_{Fe}(r) = 0$** and investigate the whole enrichment process by the BCG.

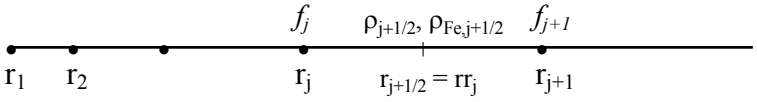
This is typical: to set up a realistic problem involves uncertainties, and subjective choices in the design of the simulation.
Results depends on these choices, too! Real problems are dirty.
Can you think to better way to start the analysis?

Let us focus on the **diffusion term** for now (which is the most problematic)

$$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial\left(r^2 D\rho\frac{\partial Z_{Fe}}{\partial r}\right)}{\partial r} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial f(\rho_{Fe})}{\partial r} \qquad (f = \text{flux})$$

Here we are **assuming that *D* = constant, ρ(r) does not change with time** and, of course, the iron density ρ_{Fe} is "inside" the abundance $Z_{Fe} = 1.4\ \rho_{Fe}/\rho$.

Note the "correct" centering of the variables on the two grids:



$$f_j = r_j^2 D\rho_j \frac{Z_{Fe,j+1/2} - Z_{Fe,j-1/2}}{r_{j+1/2} - r_{j-1/2}} \qquad \text{with} \quad \rho_j = \frac{1}{2}(\rho_{j-1/2} + \rho_{j+1/2})$$

$(\nabla Z_{Fe})_j$

$$\rightarrow \quad \left(\frac{\partial f}{\partial r}\right)_{j+1/2} = \frac{D\, r_{j+1}^2 \rho_{j+1}(\nabla Z_{Fe})_{j+1} - D\, r_j^2 \rho_j (\nabla Z_{Fe})_j}{r_{j+1} - r_j}$$

Thus, the **time integration** step is: this is a divergence

$$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial\left(r^2 D\rho\frac{\partial Z_{Fe}}{\partial r}\right)}{\partial r} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial f(\rho_{Fe})}{\partial r}$$

$=\partial(r^3/3)$

this is our unknown

$$\frac{\rho_{Fe,j+1/2}^{n+1} - \rho_{Fe,j+1/2}^{n}}{\Delta t} = \frac{1}{1.4}\frac{f_{j+1}^n - f_j^n}{(r_{j+1}^3 - r_j^3)/3}$$

Remember:

$$\text{div} f = \lim_{V\to 0}\frac{\oint_{S(V)} \mathbf{f}\cdot d\mathbf{S}}{V}$$

Where the superscript "n" refers to the **time level** (here we assume to have the solution at time "n" (= t) and want to update it at time "n+1" = t+Δt.

The time integration requires **two boundary conditions (BC)**, at the two extremes of the grid: $\rho_{Fe}(1) = \rho_{Fe}(2)$ and $\rho_{Fe}(jmax) = \rho_{Fe}(jmax-1)$

and same for Z_{Fe}. I define the "**active**" grid the points [2, jmax-1].

Important: the choice of Δt is constrained by stability reasons!

For **stability**, using the integration method above, it must be:

$$\Delta t \leq \frac{\Delta x^2}{2D}$$

what is the physical timescale associated to the lengthscale Δx?

You will learn later in the class why is that.

→ At the beginning of **every** time step, you must calculate Δt which satisfies this condition, so:

$$\Delta t = C \min\left\{\frac{\Delta x^2}{2D}\right\}$$

With C < 1, say C = 0.5

Of course, in our standard case D and Δx do not change with time, so we could calculate Δt just once, before to enter the time loop. But in the general case D can vary → allow this in the program.

The fortran code for this steps looks like this:

```
      open(20,file='kappa.dat'.status='unknown')
      vturb=260.e5    !!  as the Perseus cluster
      lturb=15.*cmkpc  !!  It is an array. Quite uncertain! Poor theory and observations
      rscala=30.*cmkpc  !!  This is used to make kappa = kappa(r).     !!
      kappa=0.11*vturb*lturb  !!  It is an array

!! do j=1,jmax     !!  This set up the values for kappa = D.  Constant or variable.
!!!     kappa(j)=0.11*vturb*lturb    !! constant !!
!!     kappa(j)=kappa(j)-0.6*kappa(j)*exp(-(r(j)/rscala)**2)
!! enddo

      n=0
1000  continue      !! here start the main time cycle
      n=n+1

!! calculate the timestep (to be modified if the grid is non-uniform)

      dt=0.4*(r(5)-r(4))**2/(2.*kappa(5))  !!
      time=time+dt
!!    print*,'n,dt (yr),time (Gyr) = ',n,real(dt/years), &
!!           real(time/1.e9/years)
```

here kappa = D!

← look!

Ok only for Δr constant

and to end the cycle:

```
      if (time.ge.tend) goto1001
      if (n.ge.ncycle) goto1001

      goto 1000

1001  continue
```

Recap: the full equations is:

$$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial (r^2 D\rho \frac{\partial Z_{Fe}}{\partial r})}{\partial r} + S_{Fe}(r,t)$$

Diffusion only: $$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial \left(r^2 D\rho \frac{\partial Z_{Fe}}{\partial r}\right)}{\partial r}$$

define first: $$\mathrm{gradzfe}_j^n = \frac{Z^n_{Fe,j+1/2} - Z^n_{Fe,j-1/2}}{r_{j+1/2} - r_{j-1/2}}$$ centered in r_j

$$\rho^{n+1}_{Fe,j+1/2} = \rho^n_{Fe,j+1/2} + \frac{\Delta t}{1.4}\frac{\left[(r^2 D\rho \cdot \mathrm{gradzfe})^n_{j+1} - (r^2 D\rho \cdot \mathrm{gradzfe})^n_j\right]}{(r^3_{j+1} - r^3_j)/3}$$

Again, look carefully at the centering of the variables.

The fortran code for this steps looks like this

```
!  diffusive step    !  check the Fe conservation !

 do j=2,jmax-1
    gradzfe(j)=(zfe(j)-zfe(j-1))/(rr(j)-rr(j-1))  !! dZ/dr centered at "j" !!
 enddo
 gradzfe(1)=0.         !! B.C. !!
 gradzfe(jmax)=0.

 do j=2,jmax-1
    rhojp1=0.5*(rho(j+1)+rho(j))  !! rho centered at "j+1" !!
    rhoj=0.5*(rho(j-1)+rho(j))    !! rho centered at "j" !!
    rhofe(j)=rhofe(j) &
            + (dt/1.4)*(r(j+1)**2*kappa(j+1)*rhojp1*gradzfe(j+1) &
            -r(j)**2*kappa(j)*rhoj*gradzfe(j))   &
             / (0.33333333*(r(j+1)**3-r(j)**3))
         zfe(j)=1.4*rhofe(j)/rho(j)  !! update Z_Fe with the new rho_Fe !!
       enddo
2000   format(3(1pe12.4))

!! set the boundary conditions (outflows)

       zfe(1)=zfe(2)
       zfe(jmax)=zfe(jmax-1)
       rhofe(1)=rhofe(2)
       rhofe(jmax)=rhofe(jmax-1)
```

We have almost everything to proceed:

1) ρ(r) is the gas density we calculated in the first part of the project

2) For the initial abundance we have **several choices**:

a) we can start with the **observed abundance** in Perseus (include the 1.4 factor!),

$$Z_{Fe}(r) = 0.3\,\frac{2.2 + (r_{kpc}/80)^3}{1 + (r_{kpc}/80)^3}\; Z_{Fe,\odot} \qquad Z_{Fe,\odot} = 1.8 \times 10^{-3}$$

and check if the BCG + diffusion maintain the observed Z_{Fe}.

b) Or we can choose to set $Z_{Fe} = Z_{Fe,out} = 0.4\, Z_{Fe,\odot}$ **or, better, $Z_{Fe} = 0$**, and check if the BCG is able to enrich the cluster and **generate the observed Z_{Fe} peak** (my preferred approach at the moment).

To test the diffusion step only, we can follow approach a) and we can also "remove" the background outer metallicity ($Z_{out} = 0.4\, Z_{Fe,\odot}$), just to isolate the Fe-peak generated by the SNIa.

$$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial (r^2 D \rho \frac{\partial Z_{Fe}}{\partial r})}{\partial r} + S_{Fe}(r,t)$$

Start by setting the initial condition for `zfe` and `rhofe`. I modified the Rebusco abundance formula as below:

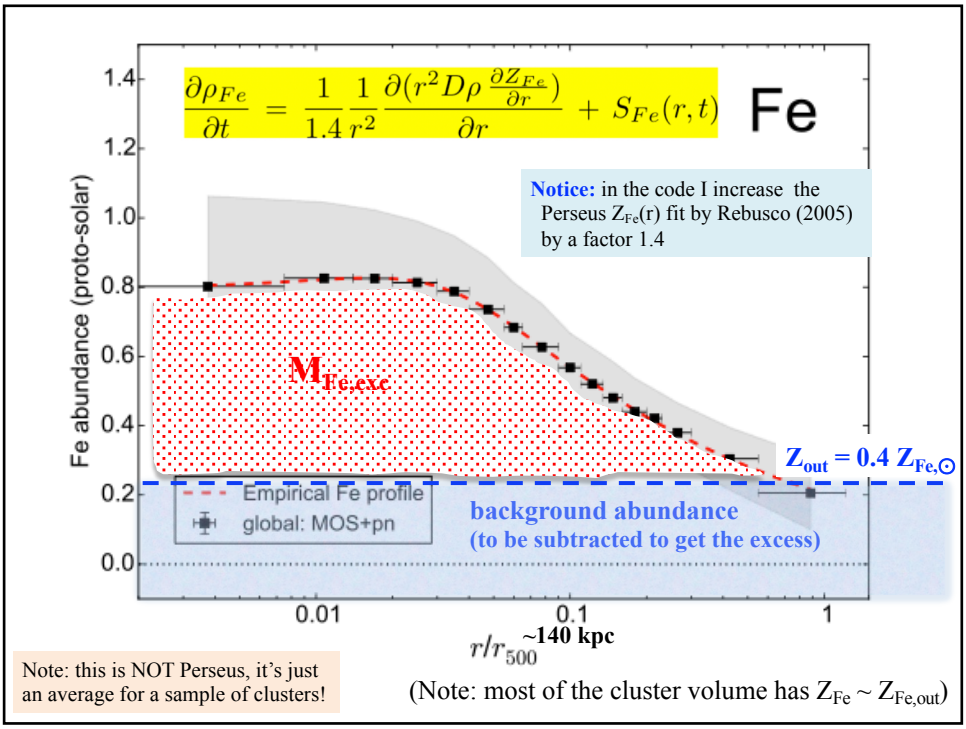
```
zfeout=0.4*zfesol    !! this is the background abundance !!

do j=1,jmax
   x=rr(j)/(80.*cmkpc)
   zfeobs(j)=zfesol*0.3*1.4*1.15*(2.2+x**3)/(1+x**3)/1.15  !Perseus!
   zfeobs(j)=zfeobs(j) - zfeout   !! subtract z_Fe,out
   zfeobs(j)=max(zfeobs(j),0.)
   zfe(j)=0. !!zfeout !!zfeobs(j)  !! which initial zfe? !!
   rhofe(j)=rho(j)*zfe(j)/1.4
   rhofeobs(j)=rho(j)*zfeobs(j)/1.4
enddo

 do j=1,jmax
    zfest(j)=1.*zfesol    !! set the stellar abundance !!
 enddo
```

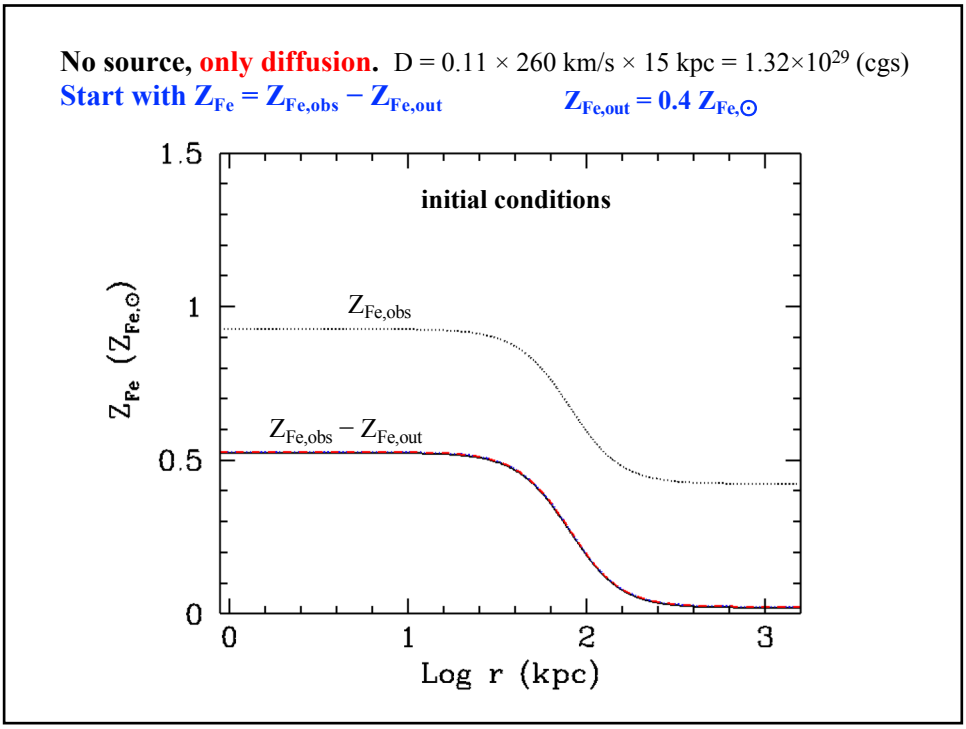
the plateau level

(plateau)

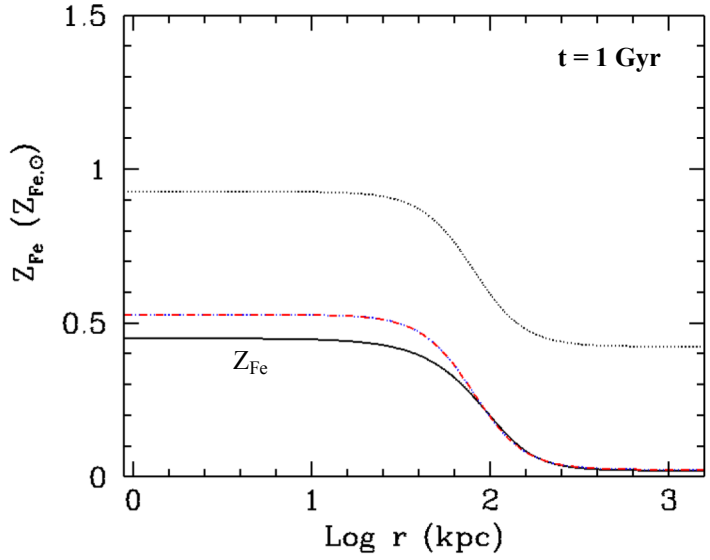
←?

This is needed for the iron source term

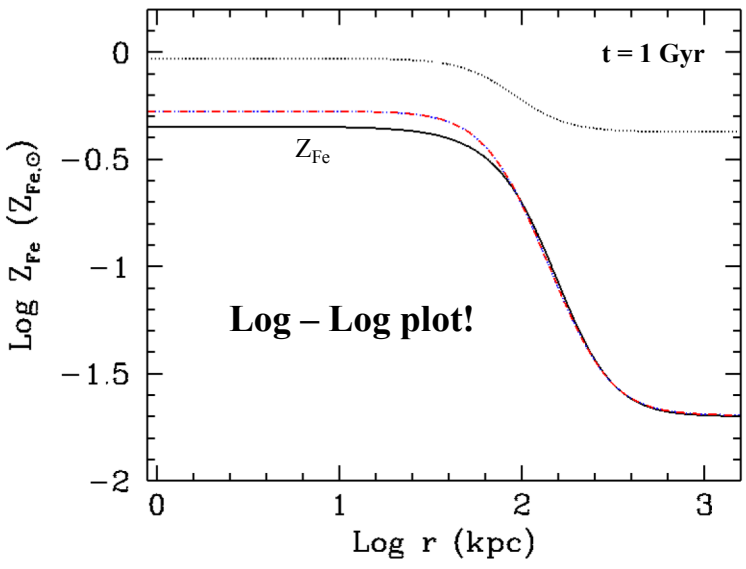

$\frac{\partial \rho_{Fe}}{\partial t} = \frac{1}{1.4}\frac{1}{r^2}\frac{\partial (r^2 D \rho \frac{\partial Z_{Fe}}{\partial r})}{\partial r} + S_{Fe}(r,t)$
Fe
Notice: in the code I increase the Perseus $Z_{Fe}(r)$ fit by Rebusco (2005) by a factor 1.4
$M_{Fe,exc}$
$Z_{out} = 0.4\ Z_{Fe,\odot}$
Empirical Fe profile
global: MOS+pn
background abundance
(to be subtracted to get the excess)
Fe abundance (proto-solar)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.01
0.1
1
r/r_{500}
~140 kpc
Note: this is NOT Perseus, it's just an average for a sample of clusters!
(Note: most of the cluster volume has $Z_{Fe} \sim Z_{Fe,out}$)


No source, only diffusion. D = 0.11 × 260 km/s × 15 kpc = 1.32×10^{29} (cgs)
Start with $Z_{Fe} = Z_{Fe,obs} - Z_{Fe,out}$
$Z_{Fe,out} = 0.4\ Z_{Fe,\odot}$
initial conditions
$Z_{Fe,obs}$
$Z_{Fe,obs} - Z_{Fe,out}$
Z_{Fe} ($Z_{Fe,\odot}$)
1.5
1
0.5
0
0
1
2
3
Log r (kpc)

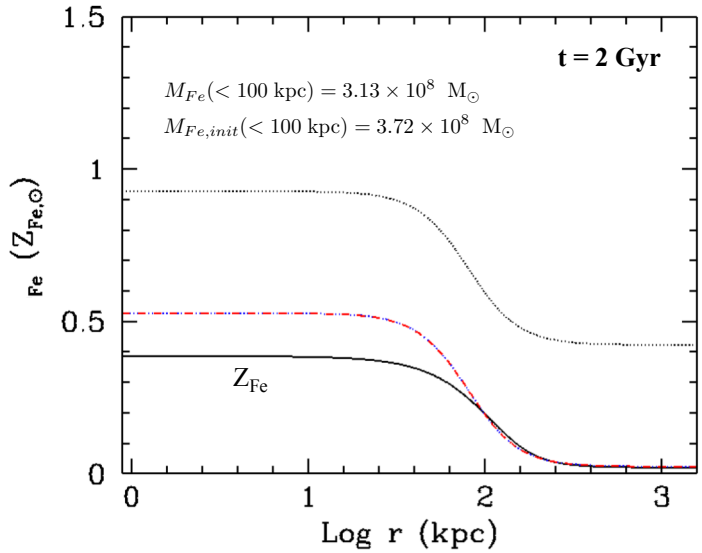
No source, only diffusion. D = 0.11 × 260 km/s × 15 kpc = 1.32×10^{29} (cgs)
Start with $Z_{Fe} = Z_{Fe,obs} - Z_{Fe,out}$



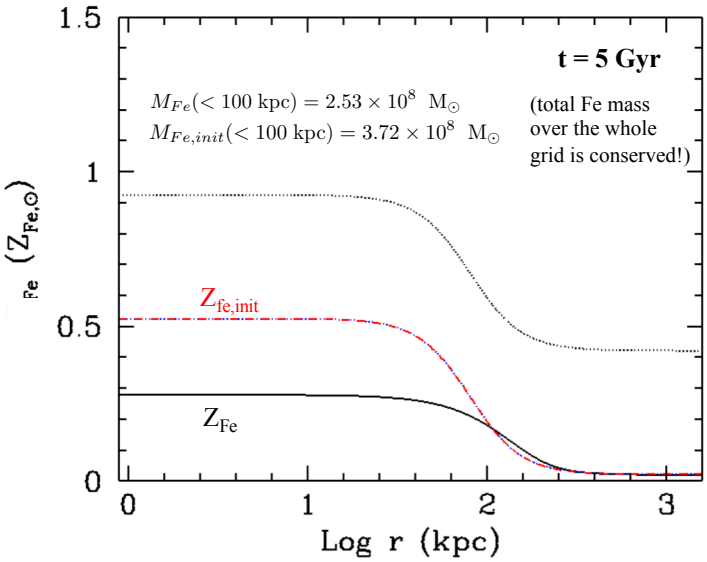
No source, only diffusion. D = 0.11 × 260 km/s × 15 kpc = 1.32×10^{29} (cgs)
Start with $Z_{Fe} = Z_{Fe,obs} - Z_{Fe,out}$



No source, only diffusion. D = 0.11 × 260 km/s × 15 kpc = 1.32×10^{29} (cgs)
Start with $Z_{Fe} = Z_{Fe,obs} - Z_{Fe,out}$

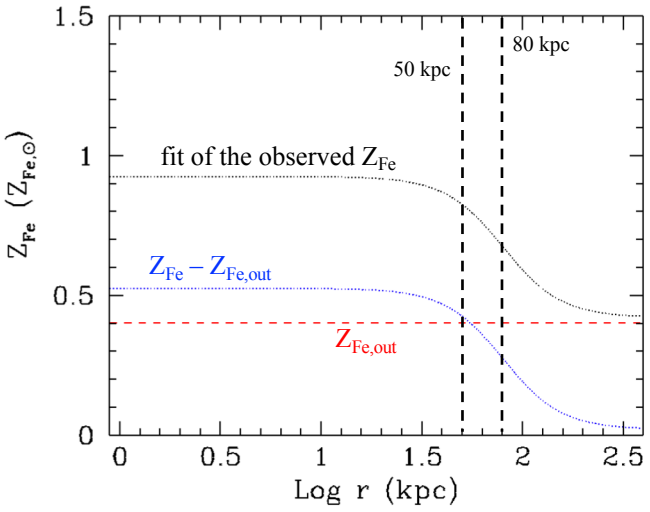


No source, only diffusion. D = 0.11 × 260 km/s × 15 kpc = 1.32×10^{29} (cgs)
Start with $Z_{Fe} = Z_{Fe,obs} - Z_{Fe,out}$



Estimate the timescale for diffusion (why?)

What is the width of the observed Fe peak?



Estimate the timescale for diffusion (why?)

$$\tau_{\rm diff} \sim \frac{L^2}{D} \sim \frac{(80\ {\rm kpc})^2}{1.3 \times 10^{29}} \sim 14.8\ {\rm Gyr}$$

But if you take L = 50 kpc → $\tau_{diff} \sim 5.8$ Gyr

With your **only-diffusion models** you can do better than this order-of-magnitude estimate, though there is not a unique way to calculate it.

For example: you can define the **diffusion timescale** as the time needed for the Fe-peak at r ~ 0 to reduce by a factor of 2 (or 3).
Even better, you can define the timescale as the time needed to reduce the iron mass M_{Fe} within some radius (50 kpc? 80 kpc? 100 kpc?) by a factor of 2 (say).
Experiment, and choose the estimate that seems more appropriate!

Last thing: the source term. Metals are injected in the ICM by:

1) **SNIa**, which explode at a certain rate. Locally, the rate must be proportional to the **stellar density** (described by the Hernquist profile we have used above for the BCG): more stars → more SNIa.

2) **stellar winds** (stars are metal rich). This source is also ∝ $\rho_*(r) = \frac{M_{BCG}}{2\pi}\frac{a}{r}\frac{1}{(r+a)^3}$

The way to express these source terms is a **bit complicated**, you do not need to know all this, but I will tell you anyway.

We **use two parameters, α_* and α_{Ia}**, defined such as: $\alpha_* M_* = \dot{M}_*$, $\alpha_{Ia} M_* = \dot{M}_{Ia}$

(α_* & α_{Ia} are the **specific mass return rates**)

stellar mass of BCG — total stellar mass loss rate — rate of total mass injection by SNIa

Stellar & SN people give us these numbers for **old** BCGs:

uncertain number, plausible range: [5 – 20 × 10^{-22}]

$\alpha_*(t_{now}) \sim 4.7 \times 10^{-20}\ \mathrm{s}^{-1}$ $\alpha_{Ia}(t_{now}) \sim 9 \times 10^{-22}\ \mathrm{s}^{-1}$

→ **SNIa contribution to the mass loss is negligible, but dominate the production of Fe!**

Indeed, both α_* and α_{Ia} **depend on time** and **they were larger in the past**. We will neglect this at the beginning by assuming the α factors constant, but we will use α(t) later. The two ‘‘alphas’’ are usually written as:

$$\alpha_*(t) = 4.7 \times 10^{-20} \left(\frac{t}{t_{now}}\right)^{-1.26}$$

$$\alpha_{Ia}(t) = 8.87 \times 10^{-22} \left(\frac{t}{t_{now}}\right)^{-s}$$

The stellar α_* depends on the details of a **single stellar population** evolution and is reasonably well known. Here t_{now} is the current cosmic time (~13.7 Gyr). We evolve the system from the **initial time of 8.7 Gyr**, up to t_{now}. That is, we are assuming that the galaxy cluster **evolve for a total time = 5 Gyr**. (This is a typical age for galaxy clusters). These numbers can be changed, of course. For example, we may prefer to evolve the cluster for 8 Gyr, from t = 5.7 Gyr to t_{now} = 13.7 Gyr.

The stars in the BCG have $Z_{Fe} \sim 1\ Z_{Fe,\odot} \sim 1.8\times10^{-3}$ →

$$S_{Fe,*} \equiv \dot{\rho}_{Fe,*} = \rho_*(r)\alpha_*(t)Z_{Fe,*}(r)/1.4$$

(assume $Z_{Fe,*}$ constant with radius)

The situation for α_{Ia} is, unfortunately, more uncertain and complicated.

Of course, α_{Ia} must depend on the **SNIa rate R_{Ia}** (number of SNIa per unit time), which depends on many things.

Astronomers like(?) to use strange units, so they measure the SN rate with "SNu", aka "**supernova unit**", defined as:

SNu = number of SNIa per $10^{10}\,L_{B,\odot}$ per century

Observational estimates of SNu for elliptical galaxies (BCGs in particular) are quite uncertain, with SNu = [0.05 – 0.5] at t = t_{now}.

Moreover, the SNIa rate depends on time, in a poorly known way. Let's assume

$$SNu(t) = SNu(t_{now})\left(\frac{t}{t_{now}}\right)^{-s} \qquad \text{with s} \sim 1.1$$

Now we need to relate SNu with α_{Ia}.
Then we need to relate α_{Ia} to the SNIa Fe source term, $S_{Ia} \equiv \dot{\rho}_{Fe,Ia}$

What is the relation between SNu and α_{Ia} and S_{Ia}?

by definition of α_{Ia}: $\alpha_{Ia} M_* = \dot{M}_{Ia}$ → $\alpha_{Ia} = \dfrac{R_{Ia} M_{Ia}}{M_*}$ s^{-1} (~ 1.4 $M_\odot$)

and by definition of SNu: $R_{Ia}(t) = \dfrac{SNu(t)\left(\frac{L_B}{10^{10} L_{B,\odot}}\right)}{3.156 \times 10^9}$ s^{-1} (B-band luminosity of the BCG; 100 yr)

$L_B = M_*/(M/L)_B$

$$\rightarrow \alpha_{Ia} = \frac{SNu\left(\frac{L_B}{10^{10} L_{B,\odot}}\right) M_{Ia}}{3.156 \times 10^9\, M_*} = 4.44 \times 10^{-20} \frac{\mathrm{SNu}}{(M/L)_B} \sim 5.92 \times 10^{-21} \mathrm{SNu} \quad \mathrm{s}^{-1}$$

(~7.5)

and, finally: (~ 0.744 $M_\odot$)

$$S_{Ia} \equiv \dot{\rho}_{Fe,Ia} = \rho_*(r)\,\alpha_{Ia}\frac{M_{Fe,Ia}}{M_{Ia}} = 3.13 \times 10^{-21} \rho_*(r)\, SNu \qquad (\mathrm{g \cdot cm^{-3} \cdot s^{-1}})$$

For a **fiducial value SNu = 0.15**: $S_{Ia} = \rho_*(r)\, 4.7 \times 10^{-22}$

(of course S_{Ia} scales linearly with SNu)

The **two α depend on time** and they were larger in the past, but neglect this for now.

How do we use the two α to get the source term (that is $\dot{\rho}_{Fe}$ to add to the ρ_{Fe} equation)? Think about it. You need to know that

1) the stellar Fe abundance is ~ 1 solar;
2) every SNIa ejects a total mass of 1.4 $M_{\odot}$;
3) every SNIa ejects 0.74 $M_{\odot}$ of iron;
[4) the SNIa rate for our BCG galaxy is $R_{Ia} \sim 0.01 - 0.1$ SNIa/yr (notice the units!)]

Try to find an expression for the source terms!

You should get something like:

contribution by stars

contribution by SNIa (for SNu=0.15)

$$S_{fe}(r) = \rho_*(r)\left[\alpha_*(t)\frac{Z_{Fe}(r)}{1.4} + \alpha_{SN}(t)\frac{M_{Fe,Ia}}{M_{Ia}}\right] \simeq \rho_*(r)\left[6\times10^{-23} + 4.7\times10^{-22}\right]$$

this can be $2 - 15\times10^{-22}$ and you should vary this number in this range and check the results

With these parameters, the **total** source term is:

$$S_{Fe}(r) = 5.3\times10^{-22}\,\rho_*(r)$$

This explains the seemingly complicated form of the source term:

contribution by stars

contribution by SNIa (for SNu=0.15)

$$S_{fe}(r) = \rho_*(r)\left[\alpha_*(t)\frac{Z_{Fe}(r)}{1.4} + \alpha_{SN}(t)\frac{M_{Fe,Ia}}{M_{Ia}}\right] \simeq \rho_*(r)\left[6\times10^{-23} + 4.7\times10^{-22}\right]$$

this can be $2 - 15\times10^{-22}$ and you should vary this number in this range and check the results

Where in the last term **we have neglected the time dependency** and assumed the values for $t = t_{now}$, that is the values for an old galaxy with no star formation.

In the program `pro1_for_stud.f90` (no more available on Virtuale… ugh!) the α are time dependent:

```
slope=1.1
alphast=4.7e-20*(time/tnow)**(-1.26)
alphasn=4.436e-20*(snu/aml)*(time/tnow)**(-slope)
```

remove this to make the SNIa source time-independent

For the Herquist stellar mass discussed in the hydrostatic equilibrium part, the **stellar density is:**

$$\rho_*(r) = \frac{M_*}{2\pi} \frac{a}{r\,(r+a)^3}$$

$$r_{1/2} = (1+\sqrt{2})a = 12 \text{ kpc}$$

$$M_* = 10^{12}\ \mathrm{M}_\odot$$

this is the total stellar mass of the central galaxy in a typical massive cluster

Roadmap of the exercise:

1) **Start with the observed Z_{Fe} profile at t = 0.**
   a) Calculate the total M_{Fe} (over the whole grid) and M_{Fe}(100 kpc)
   b) Evolve ρ_{Fe} **with diffusion only** and check how the Z_{Fe} profile changes over time **(plot it at t = 1 Gyr, 2 Gyr, 5 Gyr)**. Compare the time for "significant" (what does it mean?) **peak erasing** with the order of magnitude diffusion time.
   c) Calculate (at the three times above) M_{Fe}(100 kpc) and M_{Fe}(whole grid). Is M_{Fe} conserved?

2) **Now neglect diffusion** and **consider the source term only** and start with **Z_{Fe} = 0 everywhere.**
   a) Evolve ρ_{Fe} and check how the Fe peak is built. **Measure the width of the Fe peak** and compare with the galaxy size. How do you measure the Fe peak width?
   b) compare M_{Fe}(whole grid) with the expected $M_{Fe,theo}$ from the analytic source term.

80

(New) roadmap of the exercise:

3) **Now, diffusion and source together.** Start with $Z_{Fe} = 0$ everywhere.
   a) Evolve ρ_{Fe} and plot ρ_{Fe} profile at the three usual times (1,2,5 Gyr).
   b) Compare $Z_{Fe}(r)$ with the observed one (meaning $Z_{Fe,obs} - Z_{Fe,out}$).
   c) Vary the SNIa rate and diffusion coefficient (in a reasonable way) to get the best agreement.

(if you plan to take the "Galaxy Clusters" class, do not forget this exercise!)

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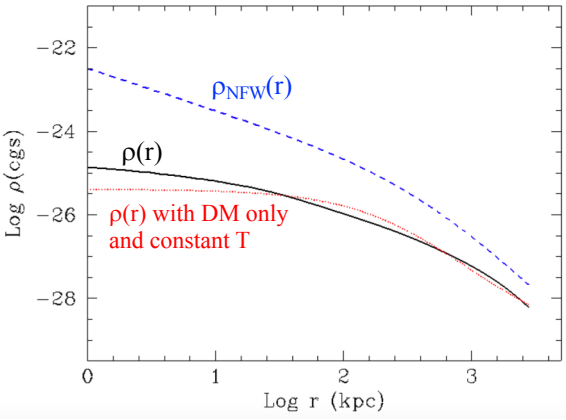
Discussion on the Fe-diffusion project

Here I briefly report my results and some astrophysical conclusions on this project. I set all my parameters to be consistent with ones in the slides, the same you should also use.

Because some result is grid-dependent, it is necessary to indicate what grid I use:

rmin = 0, rmax = 2.8 Mpc, jmax = 5000, uniform → Δr = 560 pc.



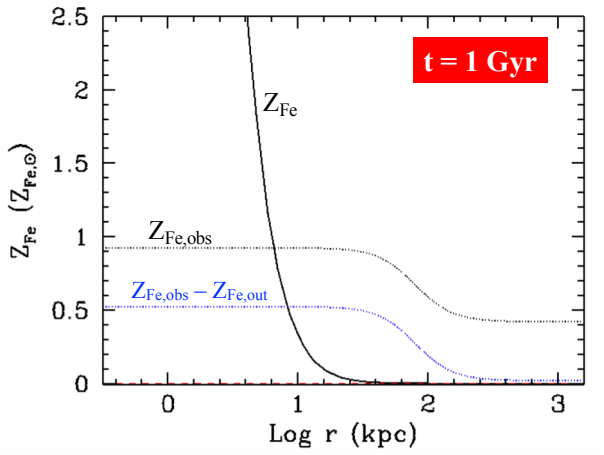
with a $\rho_0 = 1.6\times10^{-25}$ g/cm^3
[with BCG + T(r)]

I get a baryon fraction
$f_b \sim 0.169$

Effect of the Fe source term alone

$M_{\rm gas}(100\ {\rm kpc}) \sim 9.33 \times 10^{11}\ {\rm M}_\odot$

Start with Z_{Fe} = 0 everywhere
SNu = 0.15



M_{Fe} theo ~ 1.67×10^7 $M_\odot$

M_{Fe} tot ~ 1.67×10^7 $M_\odot$ (yes!)

M_{Fe}(100 kpc) ~ 1.52×10^7 $M_\odot$

Why less Fe at 100 kpc?

What is the width of the Fe-peak?

Analytic (theory) iron mass:

Source term is:

$$S_{fe}(r) = \rho_*(r)\left[\alpha_*(t)\frac{Z_{Fe}(r)}{1.4} + \alpha_{SN}(t)\frac{M_{Fe,Ia}}{M_{Ia}}\right] \simeq \rho_*(r)\left[6\times10^{-23} + 4.7\times10^{-22}\right]$$

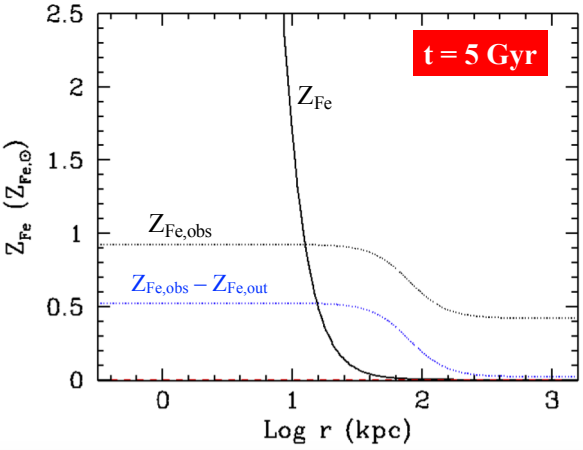
→ $$M_{\rm Fe}^{\rm theo} = 5.3\times10^{-22} M_* t$$ (check units!)

This is ok only for **time-independent sources**!

Effect of the Fe source term alone

$M_{\rm gas}(100\ {\rm kpc}) \sim 9.33 \times 10^{11}\ {\rm M}_\odot$

Start with Z_{Fe} = 0 everywhere
SNu = 0.15



M_{Fe} theo ~ 8.36×10^7 $M_\odot$

M_{Fe} tot ~ 8.33×10^7 $M_\odot$

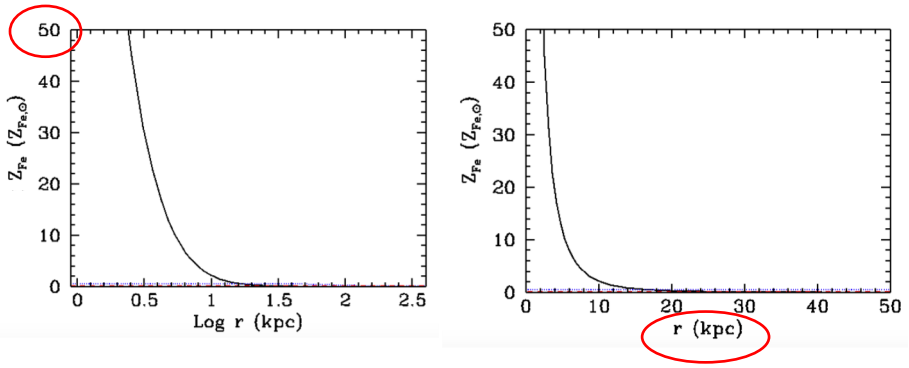
$M_{Fe}(100\ {\rm kpc}) \sim 7.58\times10^7\ M_\odot$

What is the width of the Fe-peak?

Only source, no diffusion. SNIa rate = 0.15 SNu

More plots…

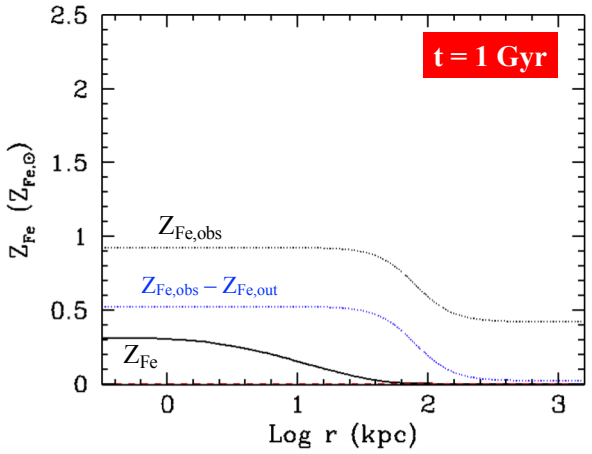
t = 5 Gyr



Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

D=K = 1.32×10^{29} (cgs units)
SNu = 0.15



M_{Fe} theo ~ 1.67×10^{7} $M_\odot$

M_{Fe} tot ~ 1.66×10^{7} $M_\odot$

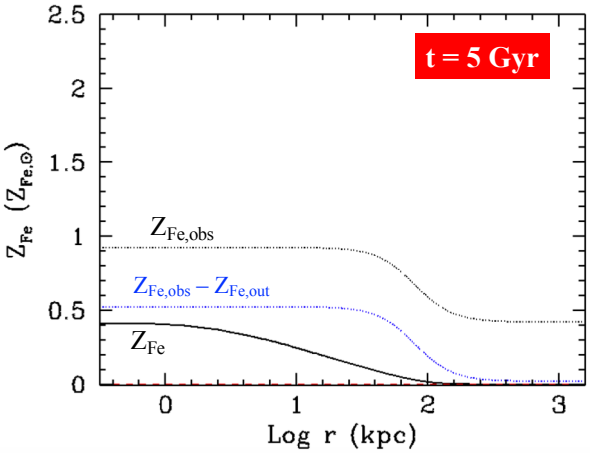
M_{Fe}(100 kpc) ~ 1.50×10^{7} $M_\odot$

M_{Fe} peak ~ 4.01×10^{8} $M_\odot$
(again for r < 100 kpc)
This is the **observed** M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

K = 1.32×10^{29} (cgs units)
SNu = 0.15



M_{Fe} theo ~ 8.36×10^{7} $M_\odot$

M_{Fe} tot ~ 8.30×10^{7} $M_\odot$

M_{Fe}(100 kpc) ~ 6.40×10^{7} $M_\odot$

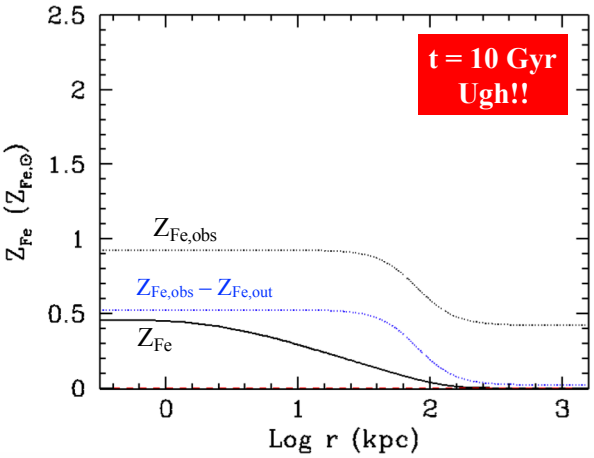
M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

$K = 1.32\times10^{29}$ (cgs units)
SNu = 0.15



M_{Fe} teorica ~ 1.67×10^{8} $M_\odot$

M_{Fe} tot ~ 1.66×10^{8} $M_\odot$

$M_{Fe}(100\ kpc) \sim 1.03\times10^{8}\ M_\odot$

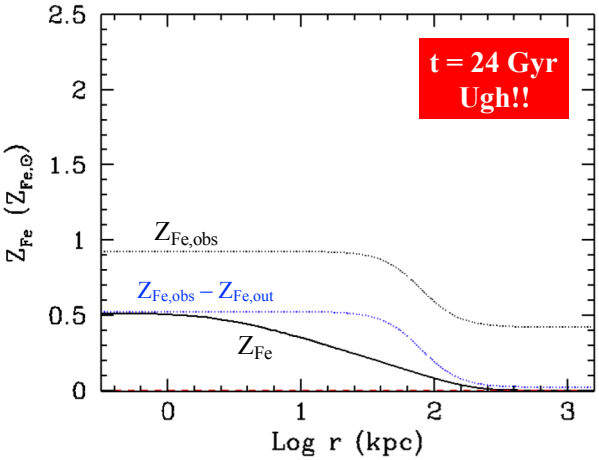
M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

$K = 1.32\times10^{29}$ (cgs units)
SNu = 0.15



M_{Fe} theo ~ 4.01×10^{8} $M_\odot$

M_{Fe} tot ~ 3.98×10^{8} $M_\odot$

$M_{Fe}(100\ kpc) \sim 1.64\times10^{8}\ M_\odot$

M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

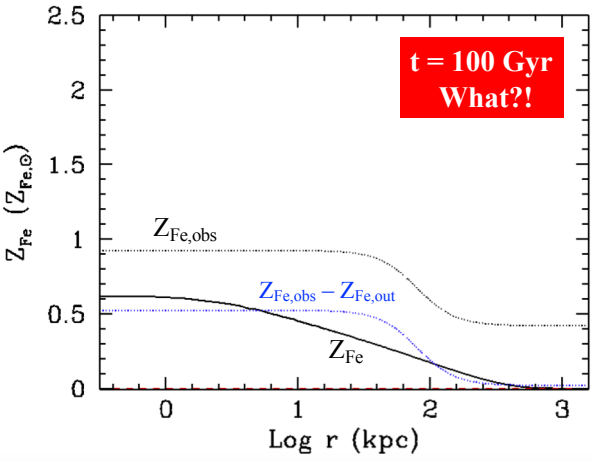
This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

Note: to increase the time is NOT the same as to increase SNu.

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

$K = 1.32\times10^{29}$ (cgs units)
SNu = 0.15



M_{Fe} theo ~ 1.67×10^{9} $M_\odot$

M_{Fe} tot ~ 1.66×10^{9} $M_\odot$

$M_{Fe}(100\ kpc) \sim 2.84\times10^{8}\ M_\odot$

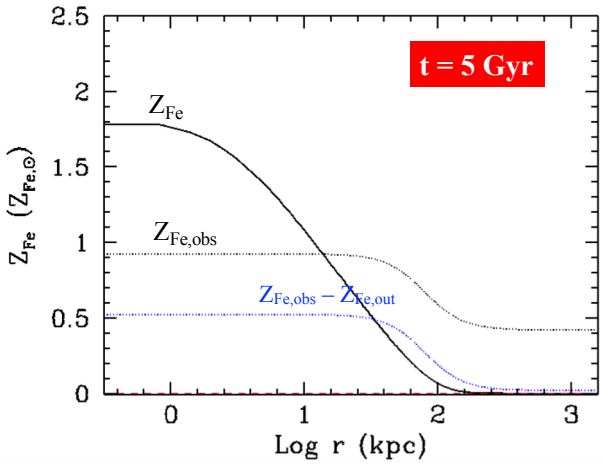
M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

$K = 1.32\times10^{29}$ (cgs units)
SNu = 0.72 (this makes all the observed Fe in 5 Gyr)



M_{Fe} theo ~ 4.01×10^{8} $M_\odot$

M_{Fe} tot ~ 3.63×10^{8} $M_\odot$

$M_{Fe}(100\ kpc) \sim 2.80\times10^{8}\ M_\odot$

M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

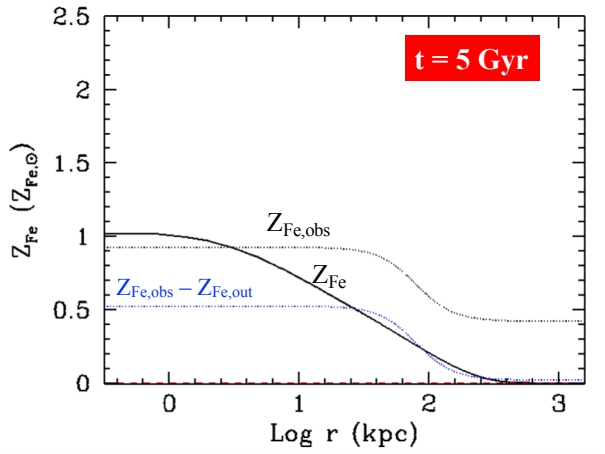
This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

Still not enough iron, and not diffused out enough…
→ try to increase both SNu and K!

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

$K = 1.06\times10^{30}$ (x8!)
SNu = 2.5 Ugh!



M_{Fe} theo ~ 1.39×10^{9} $M_\odot$

M_{Fe} tot ~ 1.24×10^{9} $M_\odot$

M_{Fe}(100 kpc) ~ 3.80×10^{8} $M_\odot$ (almost ok)

M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

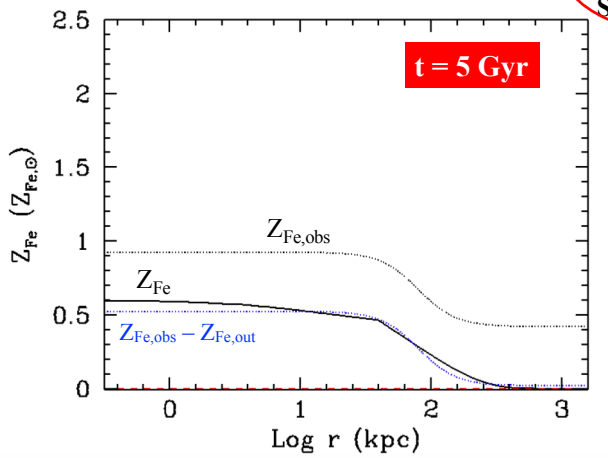
This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

~ enough iron within 100 kpc, spatial distribution not really ok…
→ try to make K = K(r)? Ex: **larger K for r ≲ 20 kpc**.

Real models: source + diffusion

Again start with $Z_{Fe} = 0$ everywhere

$K = 1.06\times10^{30}$ for r > 40 kpc
$K = 5.3\times10^{30}$ for r ≤ 40 kpc
SNu = 2.75 (const) **Ugh!**



M_{Fe} theo ~ 1.53×10^{9} $M_\odot$

M_{Fe} tot ~ 1.36×10^{9} $M_\odot$

M_{Fe}(100 kpc) ~ 4.06×10^{8} $M_\odot$ (~ ok)

M_{Fe} peak ~ 4.01×10^{8} $M_\odot$

This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$

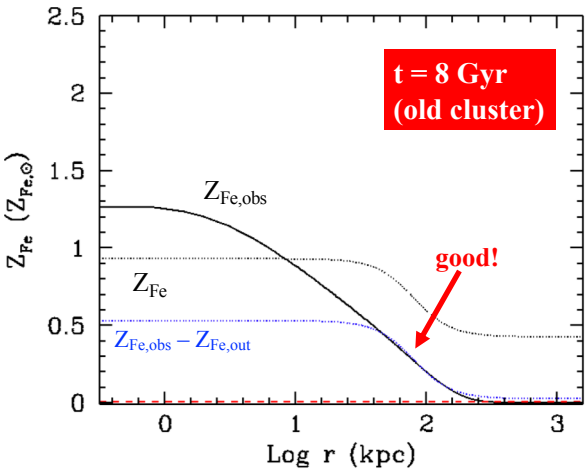
Mmh… this is a very contrived, likely wrong model!
What can we also try? **α_{Ia} as a function of time, for instance…**

Real models: source + diffusion: similar to Rebusco

Again start with $Z_{Fe} = 0$ everywhere

$K = 2.4\times10^{29}$ (cgs units)
$SNu = 0.7\times(t/t_{now})^{-1.1}$

t_{now} = 13.7 Gyr



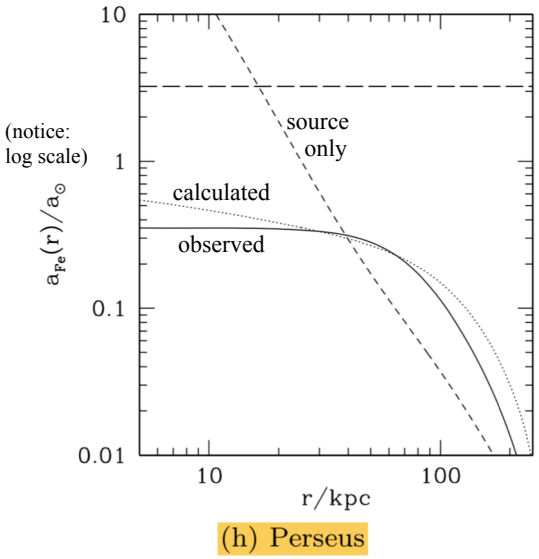
M_{Fe} theo ~ ??? $M_\odot$

M_{Fe} tot ~ 8.8×10^8 $M_\odot$

M_{Fe}(100 kpc) ~ 4.2×10^8 $M_\odot$
(~ ok)

M_{Fe} peak ~ 4.01×10^8 $M_\odot$

This is M_{Fe} calculated with $Z_{Fe,obs} - Z_{Fe,out}$



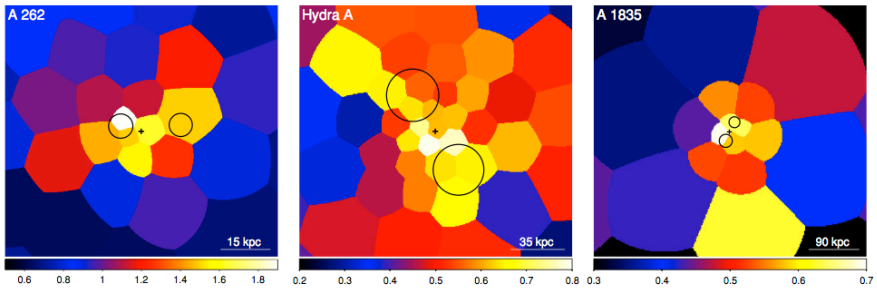
(h) Perseus

Results from
Rebusco et al. 2006

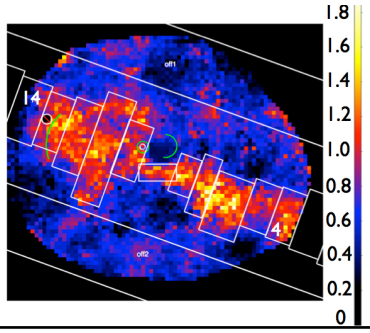
Mmh…

Real cluster abundance maps

The Astrophysical Journal Letters, 731:L23 (5pp), 2011 April 20 Kirkpatrick, McNamara, & Cavagnolo



Kirkpatrick et al., 2011



Oops!

Ettori et al., 2013

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Addendum: how can this calculation be improved?

The treatment adopted does not consider the other galaxies in the cluster, but they are the dominant component of stars in the system (~ 90% of all stars).

We can assume that the galaxies are **distributed as the dark matter**, with a density profile:

$$\rho_{gal} = \frac{\rho_{gal,0}}{\left(\frac{r}{r_s}\right)\left(1+\frac{r}{r_s}\right)^2}$$

The total mass of stars in a typical cluster is ~ 1 – 3 % of the total mass,
→ in our case this corresponds to $M_*^{tot} \sim 2 \times 10^{13}$ $\mathrm{M}_\odot$
so we can assume as a fiducial value:

$$\rho_{gal,0} = 2.2 \times 10^{-27} \quad \mathrm{g/cm^2}$$

Then we simply add a second source term similar to the one for the BCG, but multiplied by the galaxy density above.

The code on the Virtuale website

```
zfesol=1.8e-3
zfesn=0.744/1.4
snu=0.5

tnow=13.7*1.e9*years
time0=tnow-5.*1.e9*years
time=time0
```

These are needed parameters for the SNI enrichment

The idea is to evolve the model for 5 Gyr, in particular from $t_{initial}$ = 8.7 Gyr to the final time $t_{final} = t_{now}$ = 13.7 Gyr

The first part is very similar to the hydrostatic equilibrium project. The various initial gas densities `rho0` must be checked that lead to a baryon fraction `fbar~0.16`.

```
open(20,file='grv.dat')
grvnfw(1)=0.
do j=2,jmax
   grvnfw(j)=guniv*(mnfw(j)+ mhern(j))/r(j)**2   !! with BCG !!
   write(20,1002)r(j)/cmkpc,grvnfw(j)/msol
enddo
1002 format(2(1pe12.4))
close(20)
```

Notice that the code already includes the mass of the BCG `mhern`

Also, the code considers the variable temperature, by setting `tem(j)=tem2(j)`

```
open(20,file='temperature.dat',status='unknown')
 do j=1,jmax
    rkpc=rr(j)/cmkpc
    roverr0 = rkpc/r0bill
    temp1 = t0bill + tcoeffbill * roverr0**pbill
    cut=0.0
    temp2 = tmaxbill/(cut*(rkpc/60.)**2 + roverr0**qbill)
    ttt = 1./( (1./temp1)**sbill + (1./temp2)**sbill )**sinv
    tcorr = -0.12*exp(-rkpc/1.5)
    tem(j) = (ttt + tcorr)*1.e7    !! this is for NGC 5044 !!
    x=rr(j)/r500
    xx=x/0.045
    tem2(j)=ticm*1.35*(xx**1.9+0.45)/(xx**1.9+1.)* &   !! this is for Perseus !!
        1./(1.+(x/0.6)**2)**0.45
       tem(j)=tem2(j)  !!ticm --> for Perseus
    write(20,1003)rr(j)/cmkpc,tem(j),tem2(j),ticm
 enddo
close(20)
1003 format(4(1pe12.4))
```

This is NOT used. Ok for a group of galaxies.

…and the hydrostatic equation with the dT/dr term:

```
lnd(j)=lnd(j-1)-gg*(mu*mp)*(rr(j)-rr(j-1))/(boltz*temmed) &
        - (log(tem(j)) - log(tem(j-1)))
```

Here calculate gas mass `mgas` and the baryon and gas fraction

```
open(20,file='mgas.dat',status='unknown')
mgas(1)=rho(j)*4.188*r(1)**3
do j=2,jmax
   mgas(j)=mgas(j-1)+rho(j-1)*vol(j)
   write(20,1100)r(j)/cmkpc,mgas(j)/msol,mnfw(j)/msol
enddo
close(20)

!!fbar=mgas(jmax-1)/(mnfw(jmax-1)+mgas(jmax-1))
fbar=(mhern(jmax-1)+mgas(jmax-1))/(mnfw(jmax-1)+mgas(jmax-1)+mhern(jmax-1))
fgas=(mgas(jmax-1))/(mnfw(jmax-1)+mgas(jmax-1)+mhern(jmax-1))
print*,'fbari, fgas = ',real(fbar), real(fgas)

open(20,file='barfrac.dat')
do j=2,jmax-1
   fbarr(j)=(mhern(j)+mgas(j))/(mnfw(j)+mgas(j)+mhern(j))
   fgasr(j)=(mgas(j))/(mnfw(j)+mgas(j)+mhern(j))
   write(20,1100)r(j)/cmkpc,fbarr(j),fgasr(j)
enddo
close(20)
1100 format(4(1pe12.4))
```

Here calculate the `fbar` and `fgas` radial profiles (not needed, but interesting)

Now we are ready to solve the iron diffusion equation. Start by setting the initial condition for `zfe` and `rhofe`

```
zfeout=0.4*zfesol    !! this is the background abundance !!
```

the plateau level

```
do j=1,jmax
   x=rr(j)/(80.*cmkpc)
   zfeobs(j)=zfesol*0.3*1.4*1.15*(2.2+x**3)/(1+x**3)/1.15  !Perseus!
   zfeobs(j)=zfeobs(j) - zfeout    !! subtract z_Fe,out
   zfeobs(j)=max(zfeobs(j),0.)
   zfe(j)=0. !!zfeout !!zfeobs(j)  !! which initial zfe? !!
   rhofe(j)=rho(j)*zfe(j)/1.4
   rhofeobs(j)=rho(j)*zfeobs(j)/1.4
enddo
```

(plateau)

←?

```
 do j=1,jmax
    zfest(j)=1.*zfesol     !! set the stellar abundance !!
 enddo
```

This is needed for the iron source term

Calculate the iron mass, to check conservation and source term

```
!! Calculate the initial excess of iron mass

amfeiniz(1)=rhofe(1)*vol(1)
amfeiniz(1)=rhofeobs(1)*vol(1)
do j=2,jmax
   amfeiniz(j)=amfeiniz(j-1)+rhofe(j-1)*vol(j)
   amfeobs(j)=amfeobs(j-1)+rhofeobs(j-1)*vol(j)
enddo
```

Start the integration cycle:

```
      n=0
1000  continue      !! here start the main time cycle
      n=n+1

        … full cycle …

      if (time.ge.tend) goto1001
      if (n.ge.ncycle) goto1001

      goto 1000
```

Horrible! Change by using the `do while` command

Here calculate the Fe source term, `rhofedot` (SNIa + stellar winds).
Not used in the "diffusion only" experiment.

```
!! write the source terms (SNIa and stellar winds)

 slope=1.1
 alphast=4.7e-20*(time/tnow)**(-1.26)
 alphasn=4.436e-20*(snu/aml)*(time/tnow)**(-slope)

!!  print*,'alphast,sn = ',alphast,alphasn

 do j=2,jmax-2
    rhofedot(j)=(alphast*zfest(j)/1.4+alphasn*zfesn)*rhost(j)
 enddo

!!      goto776
!! source step
```

You have to skip the source term, for now (that is, remove the "!!" to the `goto` command)

…then execute the diffusion term, discussed in the last lecture.